Acta Mech 223, 2565–2590 (2012)
DOI 10.1007/s00707-012-0716-y

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Three-dimensional nonlinear thermo-elastic analysis of functionally graded cylindrical shells with piezoelectric layers by differential quadrature method

Received: 3 November 2011 / Revised: 25 January 2012 / Published online: 16 September 2012

Abstract Three-dimensional nonlinear thermo-elastic analysis of a functionally graded cylindrical shell with piezoelectric layers under the effect of asymmetric thermo-electro-mechanical loads is carried out. The strain–displacement relations are based on the nonlinear Lagrangian strain–displacement relations; that is, nonlinear terms containing derivatives of the displacement in the radial direction are included. Material properties of the shell are assumed to be graded in the radial direction according to a power law but the Poisson's ratio is assumed to be constant. Cylindrical shells are assumed to be under the effect of pressure loading in cosine form, ring pressure loads, electric and temperature fields. Numerical results of stress, displacement, electric and thermal fields are obtained by using two versions of the differential quadrature methods, namely polynomial and Fourier quadrature methods. The convergence of the solution is studied, and results of the axisymmetric loadings are verified with reported results for a cylindrical shell with material properties obeying a power law. Effects of the grading index of material properties, the temperature difference, the ratio of the mean radius to the thickness of the shell, boundary conditions, the thickness of piezoelectric layers and electric excitation on stress, displacement, electric and temperature fields are presented.

1 Introduction

Piezoelectric materials have been extensively used in various micro-electro-mechanical systems as distributed sensors and actuators for active structural control purposes. On the other hand, high load carrying capacities of laminated shells of revolution make them very attractive members for structural applications. Due to the direct and converse piezoelectric effects, piezoelectric systems produce electrical potentials or deform when subjected to mechanical pressures or placed in electric fields, respectively. Functionally graded materials (FGMs) are composites with material properties varying smoothly in one or more directions, exhibiting preferred structural responses. FGMs are usually made of two materials, that is, mixtures of metal and ceramic. The mechanical and thermal responses of materials with spatial grading in composition and microstructure can be designed for specific functions and industrial applications in order to minimize the mechanical and thermal stresses. Functionally graded structural members with piezoelectric layers are smart members that offer potential advantages in a wide range of engineering applications such as structural health monitoring, vibration and noise suppression, shape control and precision positioning. Several research works have been contributed to model and investigate the basic structural responses of piezoelectric materials, that is, in the pioneering researches of Tiersten [1].

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Wu et al. [2] used the Donnell shell theory and presented closed form solutions for a functionally graded shell with simply supported boundary conditions subjected to thermal loading. Bahtui and Eslami [3] studied the coupled thermo-elastic response of a functionally graded circular cylindrical shell subjected to thermal shock using the second-order shear deformation theory and Galerkin finite element method. Tutuncu and Temel [4] obtained axisymmetric displacements and stresses in functionally graded hollow cylinders, disks and spheres subjected to uniform internal pressure using the plane elasticity theory and complementary functions method. Shao [5] presented solutions for temperature, displacement and stress fields in a functionally graded hollow cylinder using a multilayered approach based on the theory of laminated composites. Chen and Shen [6] used the Fourier and the power series expansion methods and obtained the exact solution of an orthotropic cylindrical shell of finite length with piezoelectric layers acting as sensor and actuator subjected to axisymmetric thermo-electro-mechanical loads. Obata and Noda [7] carried out one-dimensional thermal stress analysis of a hollow circular cylinder and a hollow sphere made of functionally graded materials under steady-state condition using the perturbation method. Peng and Li [8] presented a novel method for analyzing steady-state thermal stresses in a hollow cylinder made of functionally graded material with properties that are assumed to vary arbitrarily along the radial direction. They converted the boundary value problem associated with the thermo-elastic problem to a Fredholm integral equation and solved it numerically to determine the distribution of the thermal stresses and the radial displacement. Shen and Li [9] investigated post-buckling behavior of a cross-ply laminated cylindrical shell with piezoelectric actuators subjected to the combined action of mechanical, electric and thermal loads using the boundary layer theory of shell buckling and perturbation technique. Loy et al. [10] studied vibrations of cylindrical shells made of functionally graded materials based on the Love shell theory and obtained their frequency characteristics with the help of the Rayleigh–Ritz method. A numerical analysis of a piezoelectric strip under the effect of symmetric pressure and voltage on the upper and the lower edges with traction-free boundaries using the generalized differential quadrature method was presented by Hong et al. [11].

Sheng and Wang [12] carried out thermo-elastic vibration and buckling analysis of functionally graded piezoelectric cylindrical shells. The thermo-elastic vibration and buckling characteristics were obtained by employing the Hamilton's principle and the first-order shear deformation theory. Wu and Syu [13] investigated structural behavior of functionally graded piezoelectric cylindrical shells and presented exact solutions for simply supported boundary conditions and cylindrical bending loadings, using the method of perturbation. Shakeri et al. [14] employed the Galerkin finite element method and carried out three-dimensional elasticity analysis of laminated cylinders with piezoelectric sensor and actuator layers subjected to internal pressure loadings and uniform electric excitations at their outer surfaces. Wang and Zhong [15] investigated the behavior of a finitely long circular cylindrical shell made of cylindrically orthotropic piezoelectric/piezomagnetic material under pressure loading and a uniform temperature change and obtained numerical results using the power and the Fourier series expansion methods. Alibeigloo [16] obtained a thermo-elastic solution for axisymmetric deformations of a functionally graded cylindrical shell bonded to thin piezoelectric layers, assuming a Navier type solution for the governing equations. Akbari Alashti and Khorsand [17] carried out three-dimensional linear thermo-elastic analysis of a functionally graded cylindrical shell with piezoelectric layers under the effect of asymmetric thermo-electro-mechanical loads, using the polynomial and Fourier quadrature methods.

A review of researches carried out in the past reveals that the majority of works are focused on the axisymmetric piezo-thermo-elasticity analysis of circular cylindrical shells assuming linear strain–displacement relations. To the knowledge of the authors of the present paper, few works have been presented on nonlinear asymmetric piezo-thermo-elasticity analysis of these shells. In this paper, the Fourier and polynomial differential quadrature methods are employed to carry out three-dimensional nonlinear thermo-elastic analysis of a functionally graded cylindrical shell bonded with piezoelectric layers under the effect of asymmetric thermo-electro-mechanical loads. Numerical results of displacement, stress, electric and thermal fields are obtained and compared with reported results. Effects of the grading index, the temperature difference, the ratio of the radius of the mid-plane to the thickness of the shell and boundary conditions on these parameters are investigated. Furthermore, effects of various piezoelectric parameters such as the thickness ratio of piezoelectric layers namely the actuator and the sensor layers and the electric excitation on the stress and displacement fields of the shell, in order to optimize the response of the cylindrical host shell, are investigated.

2 Governing equations

Linear constitutive equations of a piezoelectric material subjected to stress, thermal and electric fields are considered as:

$$\begin{aligned} \sigma &= C\varepsilon - e^T\Omega - \beta T, \\ D &= e\varepsilon + \eta\Omega + PT, \end{aligned} \tag{1}$$

where σ, ε, D and Ω represent the stress, strain, electric displacement and electric fields, respectively; C, e and η denote the elastic, piezoelectric and dielectric constants, respectively; β and P are the thermal modulus and pyroelectric constants, respectively, and T denotes the temperature variation of the cylindrically laminated piezoelectric shells.

$$\sigma = \{\sigma_r, \sigma_\theta, \sigma_z, \tau_{\theta z}, \tau_{rz}.\tau_{r\theta}\}^T, \quad \varepsilon = \{\varepsilon_r, \varepsilon_\theta, \varepsilon_z, \gamma_{\theta z}, \gamma_{rz}.\gamma_{r\theta}\}^T, \quad P = \{p_r, p_\theta, p_z\}^T,$$
$$\beta = \{\beta_r, \beta_\theta, \beta_z, 0, 0, 0\}^T, \quad \Omega = \{\Omega_r, \Omega_\theta, \Omega_z\}^T, \quad D = \{D_r, D_\theta, D_z\}^T,$$
$$[C] = \begin{bmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{22} & c_{23} & 0 & 0 & 0 \\ c_{13} & c_{23} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{66} \end{bmatrix}, \quad [e]^T = \begin{bmatrix} e_{11} & 0 & 0 \\ e_{21} & 0 & 0 \\ e_{31} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & e_{53} \\ 0 & e_{62} & 0 \end{bmatrix}, \quad [\eta] = \begin{bmatrix} \eta_{11} & 0 & 0 \\ 0 & \eta_{22} & 0 \\ 0 & 0 & \eta_{33} \end{bmatrix}. \tag{2}$$

The governing equations as expressed in Eq. (1) define the converse and direct piezoelectric effects. For a shell made of non-piezoelectric material, we set $e = \eta = 0$. It is assumed that the shell is made of functionally graded and isotropic material. Hence, it has only three material parameters other than the thermal conductivity, K, namely the Young's modulus, E, the Poisson's ratio, ν, and the coefficient of thermal expansion, α. The grading of material properties in the radial direction is defined by power laws in the following forms:

$$E = E_i \left(\frac{r}{a}\right)^{m_1}, \quad \alpha = \alpha_i \left(\frac{r}{a}\right)^{m_2}, \quad K = K_i \left(\frac{r}{a}\right)^{m_3}, \tag{3}$$

where E_i, α_i and K_i are the Young's modulus, the coefficient of thermal expansion and the coefficient of thermal conductivity at the inner radius of the shell, that is, at $r = a$, and m_1, m_2 and m_3, are their grading parameters in the radial direction, respectively. The material properties of the shell are assumed to be independent of the temperature field, and the Poisson's ratio is assumed to be constant throughout the shell thickness. It is also noted that piezoelectric materials with hexagonal structural symmetry exhibit transversely isotropic behavior relative to their polarization axes. The three-dimensional field equations in cylindrical coordinates, in the absence of body force, are defined as:

$$\begin{aligned} \frac{\partial \sigma_r}{\partial r} + \frac{\sigma_r - \sigma_\theta}{r} + \frac{1}{r}\frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\partial \tau_{rz}}{\partial z} &= 0, \\ \frac{1}{r}\frac{\partial \sigma_\theta}{\partial \theta} + \frac{\partial \tau_{\theta z}}{\partial z} + \frac{\partial \tau_{r\theta}}{\partial r} + 2\frac{\tau_{r\theta}}{r} &= 0, \\ \frac{\partial \sigma_z}{\partial z} + \frac{\partial \tau_{rz}}{\partial r} + \frac{1}{r}\frac{\partial \tau_{\theta z}}{\partial \theta} + \frac{1}{r}\tau_{rz} &= 0. \end{aligned} \tag{4a}$$

And the governing equation of the electrostatic charge is reported as [1]:

$$\frac{\partial D_r}{\partial r} + \frac{D_r}{r} + \frac{\partial D_\theta}{r\partial \theta} + \frac{\partial D_z}{\partial z} = 0 \tag{4b}$$

The strain–displacement relations are based on the nonlinear Lagrangian strain–displacement relations, and nonlinear terms containing derivatives of the radial displacement are included. Considering derivatives of the radial displacement with respect to all coordinates, the nonlinear strain–displacement and the electric field–electric potential relations of a piezoelectric medium are defined as:

$$
\begin{aligned}
&\varepsilon_r = \frac{\partial u}{\partial r} + \frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2, \quad \varepsilon_\theta = \frac{u}{r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{1}{2}\left(\frac{1}{r}\frac{\partial u}{\partial \theta}\right)^2, \quad \varepsilon_z = \frac{\partial w}{\partial z} + \frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2,\\
&\varepsilon_{r\theta} = \frac{1}{2}\left(\frac{1}{r}\frac{\partial u}{\partial \theta} + \frac{\partial v}{\partial r} - \frac{v}{r} + \frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial u}{\partial \theta}\right), \quad \varepsilon_{z\theta} = \frac{1}{2}\left(\frac{\partial v}{\partial z} + \frac{1}{r}\frac{\partial w}{\partial \theta} + \frac{1}{r}\frac{\partial u}{\partial \theta}\frac{\partial u}{\partial z}\right),\\
&\varepsilon_{rz} = \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} + \frac{\partial u}{\partial r}\frac{\partial u}{\partial z}\right),\\
&\Omega_r = -\frac{\partial \psi}{\partial r}, \quad \Omega_\theta = -\frac{1}{r}\frac{\partial \psi}{\partial \theta}, \quad \Omega_z = -\frac{\partial \psi}{\partial z}.
\end{aligned} \tag{5}
$$

By combining Eq. (5) with Eq. (1), the stress and the electrical displacement components are defined in terms of the displacement and the electric potential fields, their derivatives and the temperature difference of the shell:

$$
\begin{aligned}
\sigma_r &= c_{11}\left\{\frac{\partial u}{\partial r} + \frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\} + c_{12}\left\{\frac{u}{r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{1}{2}\left(\frac{1}{r}\frac{\partial u}{\partial \theta}\right)^2\right\} + c_{13}\left\{\frac{\partial w}{\partial z} + \frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\} + e_{11}\frac{\partial \psi}{\partial r} - \beta_r T,\\
\sigma_\theta &= c_{12}\left\{\frac{\partial u}{\partial r} + \frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\} + c_{22}\left\{\frac{u}{r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{1}{2}\left(\frac{1}{r}\frac{\partial u}{\partial \theta}\right)^2\right\} + c_{23}\left\{\frac{\partial w}{\partial z} + \frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\} + e_{21}\frac{\partial \psi}{\partial r} - \beta_\theta T,\\
\sigma_z &= c_{13}\left\{\frac{\partial u}{\partial r} + \frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\} + c_{23}\left\{\frac{u}{r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{1}{2}\left(\frac{1}{r}\frac{\partial u}{\partial \theta}\right)^2\right\} + c_{33}\left\{\frac{\partial w}{\partial z} + \frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\} + e_{31}\frac{\partial \psi}{\partial r} - \beta_z T,\\
\tau_{\theta z} &= c_{44}\left\{\frac{\partial v}{\partial z} + \frac{1}{r}\frac{\partial w}{\partial \theta} + \frac{1}{r}\frac{\partial u}{\partial \theta}\frac{\partial u}{\partial z}\right\}, \qquad \tau_{rz} = c_{55}\left\{\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} + \frac{\partial u}{\partial r}\frac{\partial u}{\partial z}\right\} + e_{53}\frac{\partial \psi}{\partial z},\\
\tau_{r\theta} &= c_{66}\left\{\frac{1}{r}\frac{\partial u}{\partial \theta} + \frac{\partial v}{\partial r} - \frac{v}{r} + \frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial u}{\partial \theta}\right\} + \frac{e_{62}}{r}\frac{\partial \psi}{\partial \theta},\\
D_r &= e_{11}\left\{\frac{\partial u}{\partial r} + \frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\} + e_{21}\left\{\frac{u}{r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{1}{2}\left(\frac{1}{r}\frac{\partial u}{\partial \theta}\right)^2\right\} + e_{31}\left\{\frac{\partial w}{\partial z} + \frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\} - \eta_{11}\frac{\partial \psi}{\partial r} + p_r T,\\
D_\theta &= e_{62}\left\{\frac{1}{r}\frac{\partial u}{\partial \theta} + \frac{\partial v}{\partial r} - \frac{v}{r} + \frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial u}{\partial \theta}\right\} - \frac{\eta_{22}}{r}\frac{\partial \psi}{\partial \theta} + p_\theta T, D_z = e_{53}\left\{\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} + \frac{\partial u}{\partial r}\frac{\partial u}{\partial z}\right\} - \eta_{33}\frac{\partial \psi}{\partial z} + p_z T.
\end{aligned} \tag{6}
$$

By substituting the above equations into Eq. (4), the governing equations of equilibrium are obtained in terms of displacement fields, electric potentials and temperature of each layer of the cylindrical shell. The piezoelectric field equations are found to be:

$$
\begin{aligned}
&c_{11}\left\{\frac{\partial^2 u}{\partial r^2} + \frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r^2}\right\} + c_{12}\left\{\frac{1}{r}\frac{\partial u}{\partial r} - \frac{1}{r^2}u - \frac{1}{r^2}\frac{\partial v}{\partial \theta} + \frac{1}{r}\frac{\partial^2 v}{\partial r \partial \theta} - \frac{1}{r^3}\left(\frac{\partial u}{\partial \theta}\right)^2 + \frac{1}{r^2}\frac{\partial u}{\partial \theta}\frac{\partial^2 u}{\partial r \partial \theta}\right\}\\
&\quad + c_{13}\left\{\frac{\partial^2 w}{\partial r \partial z} + \frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial r \partial z}\right\} + e_{11}\frac{\partial^2 \psi}{\partial r^2} - \beta_r\frac{\partial T}{\partial r} + \frac{1}{r}\left\{c_{11}\left\{\frac{\partial u}{\partial r} + \frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\} + c_{12}\left\{\frac{u}{r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{1}{2r^2}\left(\frac{\partial u}{\partial \theta}\right)^2\right\}\right.\\
&\quad + c_{13}\left\{\frac{\partial w}{\partial z} + \frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\} + e_{11}\frac{\partial \psi}{\partial r} - \beta_r T - c_{12}\left\{\frac{\partial u}{\partial r} + \frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\} - c_{22}\left\{\frac{1}{r}u + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{1}{2r^2}\left(\frac{\partial u}{\partial \theta}\right)^2\right\}\\
&\quad \left. - c_{23}\left\{\frac{\partial w}{\partial z} + \frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\} - e_{21}\frac{\partial \psi}{\partial r} + \beta_\theta T\right\} + \frac{1}{r}\left\{c_{66}\left\{\frac{1}{r}\frac{\partial^2 v}{\partial \theta^2} + \frac{\partial^2 v}{\partial r \partial \theta} - \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{1}{r}\frac{\partial^2 u}{\partial r \partial \theta}\frac{\partial u}{\partial \theta} + \frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial \theta^2}\right\}\right.\\
&\quad \left. + \frac{e_{62}}{r}\frac{\partial^2 \psi}{\partial \theta^2}\right\} + c_{55}\left\{\frac{\partial^2 u}{\partial z^2} + \frac{\partial^2 w}{\partial r \partial z} + \frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial r \partial z} + \frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial z^2}\right\} + e_{53}\frac{\partial^2 \psi}{\partial z^2} = 0,
\end{aligned} \tag{7a}
$$

$$\frac{1}{r}\left\{c_{12}\left\{\frac{\partial^2 u}{\partial r\partial\theta}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r\partial\theta}\right\}+c_{22}\left\{\frac{1}{r}\frac{\partial u}{\partial\theta}+\frac{1}{r}\frac{\partial^2 v}{\partial\theta^2}+\frac{1}{r^2}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial\theta^2}\right\}+c_{23}\left\{\frac{\partial^2 w}{\partial\theta\partial z}+\frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial\theta\partial z}\right\}+e_{21}\frac{\partial^2\psi}{\partial r\partial\theta}\right\}$$
$$+c_{44}\left\{\frac{\partial^2 v}{\partial z^2}+\frac{1}{r}\frac{\partial^2 w}{\partial\theta\partial z}+\frac{1}{r}\frac{\partial^2 u}{\partial\theta\partial z}\frac{\partial u}{\partial z}+\frac{1}{r}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial z^2}\right\}+c_{66}\left\{-\frac{1}{r^2}\frac{\partial u}{\partial\theta}+\frac{1}{r}\frac{\partial^2 u}{\partial r\partial\theta}+\frac{\partial^2 v}{\partial r^2}-\frac{1}{r}\frac{\partial v}{\partial r}+\frac{v}{r^2}\right.$$
$$\left.-\frac{1}{r^2}\frac{\partial u}{\partial r}\frac{\partial u}{\partial\theta}+\frac{1}{r}\frac{\partial^2 u}{\partial r^2}\frac{\partial u}{\partial\theta}+\frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r\partial\theta}\right\}-\frac{e_{62}}{r^2}\frac{\partial\psi}{\partial\theta}+\frac{e_{62}}{r}\frac{\partial^2\psi}{\partial r\partial\theta}+\frac{2}{r}\left\{c_{66}\left\{\frac{1}{r}\frac{\partial u}{\partial\theta}+\frac{\partial v}{\partial r}-\frac{1}{r}v+\frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial u}{\partial\theta}\right\}\right.$$
$$\left.+\frac{e_{62}}{r}\frac{\partial\psi}{\partial\theta}\right\}=0, \tag{7b}$$

$$c_{13}\left\{\frac{\partial^2 u}{\partial r\partial z}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r\partial z}\right\}+c_{23}\left\{\frac{1}{r}\frac{\partial u}{\partial z}+\frac{1}{r}\frac{\partial^2 v}{\partial\theta\partial z}+\frac{1}{r^2}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial\theta\partial z}\right\}+c_{33}\left\{\frac{\partial^2 w}{\partial z^2}+\frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial z^2}\right\}$$
$$+e_{31}\frac{\partial^2\psi}{\partial r\partial z}-\beta_z\frac{\partial T}{\partial z}+c_{55}\left\{\frac{\partial^2 u}{\partial r\partial z}+\frac{\partial^2 w}{\partial r^2}+\frac{\partial^2 u}{\partial r^2}\frac{\partial u}{\partial z}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r\partial z}\right\}+e_{53}\frac{\partial^2\psi}{\partial r\partial z}$$
$$+\frac{c_{44}}{r}\left\{\frac{\partial^2 v}{\partial\theta\partial z}+\frac{1}{r}\frac{\partial^2 w}{\partial\theta^2}+\frac{1}{r}\frac{\partial^2 u}{\partial\theta^2}\frac{\partial u}{\partial z}+\frac{1}{r}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial\theta\partial z}\right\}+\frac{c_{55}}{r}\left\{\frac{\partial u}{\partial z}+\frac{\partial w}{\partial r}+\frac{\partial u}{\partial r}\frac{\partial u}{\partial z}\right\}+\frac{e_{53}}{r}\frac{\partial\psi}{\partial z}=0, \tag{7c}$$

$$e_{11}\left\{\frac{\partial^2 u}{\partial r^2}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r^2}\right\}+e_{21}\left\{\frac{1}{r}\frac{\partial u}{\partial r}-\frac{1}{r^2}u-\frac{1}{r^2}\frac{\partial v}{\partial\theta}+\frac{1}{r}\frac{\partial^2 v}{\partial r\partial\theta}-\frac{1}{r^3}\left(\frac{\partial u}{\partial\theta}\right)^2+\frac{1}{r^2}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial r\partial\theta}\right\}$$
$$+e_{31}\left\{\frac{\partial^2 w}{\partial r\partial z}+\frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial r\partial z}\right\}-\eta_{11}\frac{\partial^2\psi}{\partial r^2}+p_r\frac{\partial T}{\partial r}+\frac{1}{r}\left\{e_{11}\left\{\frac{\partial u}{\partial r}+\frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\}+e_{21}\left\{\frac{1}{r}u+\frac{1}{r}\frac{\partial v}{\partial\theta}+\frac{1}{2r^2}\left(\frac{\partial u}{\partial\theta}\right)^2\right\}\right.$$
$$\left.+e_{31}\left\{\frac{\partial w}{\partial z}+\frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\}-\eta_{11}\frac{\partial\psi}{\partial r}+p_rT\right\}+\frac{1}{r}\left\{e_{62}\left\{\frac{1}{r}\frac{\partial^2 u}{\partial\theta^2}+\frac{\partial^2 v}{\partial r\partial\theta}-\frac{1}{r}\frac{\partial v}{\partial\theta}+\frac{1}{r}\frac{\partial^2 u}{\partial r\partial\theta}\frac{\partial u}{\partial\theta}\right.\right.$$
$$\left.\left.+\frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial\theta^2}\right\}-\frac{\eta_{22}}{r}\frac{\partial^2\psi}{\partial\theta^2}\right\}+e_{53}\left\{\frac{\partial^2 u}{\partial z^2}+\frac{\partial^2 w}{\partial r\partial z}+\frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial r\partial z}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial z^2}\right\}-\eta_{33}\frac{\partial^2\psi}{\partial z^2}+p_z\frac{\partial T}{\partial z}=0. \tag{7d}$$

The field equations corresponding to the isotropic shell are defined as:

$$\frac{E}{(1+\upsilon)(1-2\upsilon)}\left\{(1-\upsilon)\left\{\frac{\partial^2 u}{\partial r^2}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r^2}\right\}+\upsilon\left\{\frac{1}{r}\frac{\partial u}{\partial r}-\frac{1}{r^2}u-\frac{1}{r^2}\frac{\partial v}{\partial\theta}+\frac{1}{r}\frac{\partial^2 v}{\partial r\partial\theta}-\frac{1}{r^3}\left(\frac{\partial u}{\partial\theta}\right)^2+\frac{1}{r^2}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial r\partial\theta}\right\}\right.$$
$$\left.+\upsilon\left\{\frac{\partial^2 w}{\partial r\partial z}+\frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial r\partial z}\right\}\right\}-\frac{1}{(1-2\upsilon)}\frac{\partial(E\alpha T)}{\partial r}+\frac{\partial E/\partial r}{(1+\upsilon)(1-2\upsilon)}\{(1-\upsilon)\left\{\frac{\partial u}{\partial r}+\frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\}$$
$$+\upsilon\left\{\frac{u}{r}+\frac{1}{r}\frac{\partial v}{\partial\theta}-\frac{1}{2r^2}\left(\frac{\partial u}{\partial\theta}\right)^2\right\}+\upsilon\left\{\frac{\partial w}{\partial z}+\frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\}+\frac{E}{(1+\upsilon)(1-2\upsilon)r}\left\{(1-\upsilon)\left\{\frac{\partial u}{\partial r}+\frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\}\right.$$
$$\left.+\upsilon\left\{\frac{1}{r}u+\frac{1}{r}\frac{\partial v}{\partial\theta}+\frac{1}{2r^2}\left(\frac{\partial u}{\partial\theta}\right)^2\right\}+\upsilon\left\{\frac{\partial w}{\partial z}+\frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\}\right\}$$
$$-\frac{E}{(1+\upsilon)(1-2\upsilon)r}\left\{(1-\upsilon)\left\{\frac{1}{r}u+\frac{1}{r}\frac{\partial v}{\partial\theta}+\frac{1}{2r^2}\left(\frac{\partial u}{\partial\theta}\right)^2\right\}\right.$$
$$\left.+\upsilon\left\{\frac{\partial u}{\partial r}+\frac{1}{2}\left(\frac{\partial u}{\partial r}\right)^2\right\}+\upsilon\left\{\frac{\partial w}{\partial z}+\frac{1}{2}\left(\frac{\partial u}{\partial z}\right)^2\right\}\right\}+\frac{E}{2(1+\upsilon)r}\left\{\frac{1}{r}\frac{\partial^2 u}{\partial\theta^2}+\frac{\partial^2 v}{\partial r\partial\theta}-\frac{1}{r}\frac{\partial v}{\partial\theta}+\frac{1}{r}\frac{\partial^2 u}{\partial r\partial\theta}\frac{\partial u}{\partial\theta}\right.$$
$$\left.+\frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial\theta^2}\right\}+\frac{E}{2(1+\upsilon)}\left\{\frac{\partial^2 u}{\partial z^2}+\frac{\partial^2 w}{\partial r\partial z}+\frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial r\partial z}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial z^2}\right\}=0, \tag{8a}$$

$$\frac{E}{(1+\upsilon)(1-2\upsilon)r}\left\{(1-\upsilon)\left\{\frac{1}{r}\frac{\partial u}{\partial\theta}+\frac{1}{r}\frac{\partial^2 v}{\partial\theta^2}+\frac{1}{r^2}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial\theta^2}\right\}+\upsilon\left\{\frac{\partial^2 u}{\partial r\partial\theta}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r\partial\theta}\right\}+\upsilon\left\{\frac{\partial^2 w}{\partial\theta\partial z}+\frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial\theta\partial z}\right\}\right\}$$
$$+\frac{E}{2(1+\upsilon)}\left\{\frac{\partial^2 v}{\partial z^2}+\frac{1}{r}\frac{\partial^2 w}{\partial\theta\partial z}+\frac{1}{r}\frac{\partial^2 u}{\partial\theta\partial z}\frac{\partial u}{\partial z}+\frac{1}{r}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial z^2}\right\}+\frac{E}{2(1+\upsilon)}\left\{-\frac{1}{r^2}\frac{\partial u}{\partial\theta}+\frac{1}{r}\frac{\partial^2 u}{\partial r\partial\theta}+\frac{\partial^2 v}{\partial r^2}-\frac{1}{r}\frac{\partial v}{\partial r}+\frac{1}{r^2}v\right.$$
$$\left.-\frac{1}{r^2}\frac{\partial u}{\partial r}\frac{\partial u}{\partial\theta}+\frac{1}{r}\frac{\partial^2 u}{\partial r^2}\frac{\partial u}{\partial\theta}+\frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r\partial\theta}\right\}+\frac{\partial E/\partial r}{2(1+\upsilon)}\left\{\frac{1}{r}\frac{\partial u}{\partial\theta}+\frac{\partial v}{\partial r}-\frac{v}{r}+\frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial u}{\partial\theta}\right\}$$
$$+\frac{E}{(1+\upsilon)r}\left\{\frac{1}{r}\frac{\partial u}{\partial\theta}+\frac{\partial v}{\partial r}-\frac{1}{r}v+\frac{1}{r}\frac{\partial u}{\partial r}\frac{\partial u}{\partial\theta}\right\}=0, \tag{8b}$$

$$\frac{E}{(1+\upsilon)(1-2\upsilon)}\left\{(1-\upsilon)\left\{\frac{\partial^2 w}{\partial z^2}+\frac{\partial u}{\partial z}\frac{\partial^2 u}{\partial z^2}\right\}+\upsilon\left\{\frac{\partial^2 u}{\partial r\partial z}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r\partial z}\right\}+\upsilon\left\{\frac{1}{r}\frac{\partial u}{\partial z}+\frac{1}{r}\frac{\partial^2 v}{\partial\theta\partial z}+\frac{1}{r^2}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial\theta\partial z}\right\}\right\}$$
$$-\frac{E\alpha}{(1-2\upsilon)}\frac{\partial T}{\partial z}+\frac{E}{2(1+\upsilon)}\left\{\frac{\partial^2 u}{\partial r\partial z}+\frac{\partial^2 w}{\partial r^2}+\frac{\partial^2 u}{\partial r^2}\frac{\partial u}{\partial z}+\frac{\partial u}{\partial r}\frac{\partial^2 u}{\partial r\partial z}\right\}+\frac{\partial E/\partial r}{2(1+\upsilon)}\left\{\frac{\partial u}{\partial z}+\frac{\partial w}{\partial r}+\frac{\partial u}{\partial r}\frac{\partial u}{\partial z}\right\}$$
$$+\frac{E}{2(1+\upsilon)r}\left\{\frac{\partial^2 v}{\partial\theta\partial z}+\frac{1}{r}\frac{\partial^2 w}{\partial\theta^2}+\frac{1}{r}\frac{\partial^2 u}{\partial\theta^2}\frac{\partial u}{\partial z}+\frac{1}{r}\frac{\partial u}{\partial\theta}\frac{\partial^2 u}{\partial\theta\partial z}\right\}+\frac{E}{2(1+\upsilon)r}\left\{\frac{\partial u}{\partial z}+\frac{\partial w}{\partial r}+\frac{\partial u}{\partial r}\frac{\partial u}{\partial z}\right\}=0. \tag{8c}$$

In the present work, a steady-state heat conduction without internal heat source is considered; hence, the governing equation of the steady-state temperature field is defined as:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(rK_r\frac{\partial T}{\partial r}\right)+\frac{\partial}{\partial z}\left(K_z\frac{\partial T}{\partial z}\right)=0. \tag{9}$$

3 Boundary conditions

Mechanical and electrical boundary conditions at the ends of the shell for two cases of simply supported and clamped boundary conditions are described by the following relations:

$$\text{Clamped:}\quad \text{at } z=0, L:\quad u=v=w=\psi=T=0 \tag{10a}$$
$$\text{Simply supported:}\quad \text{at } z=0, L:\quad u=v=\sigma_z=\psi=T=0 \tag{10b}$$

where L is the length of the shell. For a laminated shell, the interface equilibrium and compatibility conditions which must be met at interfaces of each adjacent layers, that is, between the kth and the $(k+1)$th interfaces, are expressed as:

$$\sigma_r|_k=\sigma_r|_{k+1},\ \tau_{rz}|_k=\tau_{rz}|_{k+1},\ \tau_{r\theta}|_k=\tau_{r\theta}|_{k+1},\ u|_k=u|_{k+1},\ v|_k=v|_{k+1},$$
$$w|_k=w|_{k+1},\ D_r|_k=D_r|_{k+1},\ \psi|_k=\psi|_{k+1},\ T|_k=T|_{k+1},\ K\left.\frac{\partial T}{\partial r}\right|_k=K\left.\frac{\partial T}{\partial r}\right|_{k+1}. \tag{11}$$

The surface boundary conditions of the outer piezoelectric shell that is considered as the sensor is written as:

$$\text{at}\quad r=b\text{:}\ \sigma_r=\tau_{rz}=\tau_{r\theta}=D_r=0,\quad T=0. \tag{12a}$$

Surface boundary conditions of the inner shell are as follows:

$$\text{at}\quad r=a\text{:}\quad \sigma_r=P_0(\theta,z),\quad \tau_{rz}=\tau_{r\theta}=0,\quad T=T_a. \tag{12b}$$

4 Discrete equations

In this study, two types of differential quadratures are employed to approximate the first- and the second-order derivatives of functions and discretize the governing partial differential equations into series forms. The polynomial expansion-based differential quadrature is used for the radial and the longitudinal directions and the Fourier expansion-based differential quadrature for the circumferential direction.

4.1 Fourier expansion-based differential quadrature (FDQ)

It is supposed that the desired function, that is, $f(x)$, is approximated by a Fourier series expansion of the form

$$f(x) = c_0 + \sum_{k=1}^{\frac{N}{2}} (c_k \cos kx + d_k \sin kx), \tag{13}$$

where the function $f(x)$ constitutes an $N+1$ dimensional linear vector space V_{N+1}. Using these sets of base vectors, explicit formulations to compute the weighting coefficients of the first- and the second-order derivatives of the following forms are obtained [18]:

$$\begin{aligned}
&a_{ij} = \frac{q(x_i)}{2\sin\left(\frac{x_i - x_j}{2}\right) q(x_j)} \quad i \neq j, \quad q(x_i) = \prod_{k=0}^{N} \sin\left(\frac{x_i - x_k}{2}\right), \quad a_{ii} = -\sum_{\substack{j=0\\ i\neq j}}^{N} a_{ij},\\
&b_{ij} = a_{ij}\left(2a_{ii} - \cot\left(\frac{x_i - x_j}{2}\right)\right) \quad i \neq j, \quad b_{ii} = -\sum_{\substack{j=0\\ i\neq j}}^{N} b_{ij},\\
&c_{ij} = a_{ij}\left(\frac{1}{2} + 3b_{ii}\right) - \frac{3}{2} b_{ij}\cot\left(\frac{x_i - x_j}{2}\right) \quad i \neq j, \quad \mathrm{c}_{ii} = -\sum_{\substack{j=0\\ i\neq j}}^{N} c_{ij},
\end{aligned} \tag{14}$$

where a_{ij}, b_{ij} and c_{ij} denote the weighting coefficients of the first-, the second- and the third-order derivatives of the function $f(x)$.

4.2 Polynomial-based differential quadrature (PDQ)

Several attempts have been made by researchers to develop polynomial-based differential quadrature methods. One of the most useful approaches is the one that uses the following Lagrange interpolation polynomials as test functions:

$$g_k(x) = \frac{M(x)}{(x - x_k) M^{(1)}(x_k)}, \quad k = 1, 2, \ldots, N, \tag{15a}$$

where

$$M(x) = (x - x_1)(x - x_2)\cdots(x - x_N), \quad M^{(1)}(x_i) = \prod_{\substack{k=1\\ k\neq i}}^{N} (x_i - x_k). \tag{15b}$$

Applying the above equation at N grid points, the following algebraic formulations to compute the weighting coefficients are developed:

$$a_{ij} = \frac{1}{x_j - x_i} \prod_{\substack{k=1\\ k\neq i,j}}^{N} \frac{x_i - x_k}{x_j - x_k}, \quad i \neq j; \quad a_{ii} = \sum_{\substack{k=1\\ k\neq i}}^{N} \frac{1}{x_i - x_k}. \tag{16}$$

4.3 Three-dimensional differential quadratures

The above differential quadrature approximation corresponds to a one-dimensional formulation. The differential quadrature approximation can be easily extended from the above formulation to other coordinates. The first-order derivatives in the three-dimensional formulation are approximated by:

$$\left.\frac{\partial f}{\partial r}\right)_{ijk} \approx \sum_{l=1}^{N} A_{il}^{(1)} f_{ljk}, \quad \left.\frac{\partial f}{\partial \theta}\right)_{ijk} \approx \sum_{m=1}^{M} B_{jm}^{(1)} f_{imk}, \quad \left.\frac{\partial f}{\partial z}\right)_{ijk} \approx \sum_{n=1}^{P} C_{kn}^{(1)} f_{ijn}, \tag{17a}$$

and the second-order derivatives can be approximated by:

$$\begin{aligned}
&\left.\frac{\partial^2 f}{\partial r^2}\right)_{ijk} \approx \sum_{l=1}^{N} A_{il}^{(2)} f_{ljk}, \ \left.\frac{\partial^2 f}{\partial \theta^2}\right)_{ijk} \approx \sum_{m=1}^{M} B_{jm}^{(2)} f_{imk},\\
&\left.\frac{\partial^2 f}{\partial z^2}\right)_{ijk} \approx \sum_{n=1}^{P} C_{kn}^{(2)} f_{ijn}, \ \left.\frac{\partial^2 f}{\partial r \partial \theta}\right)_{ijk} \approx \sum_{l=1}^{N} A_{il}^{(1)} \sum_{m=1}^{M} B_{jm}^{(1)} f_{lmk},\\
&\left.\frac{\partial^2 f}{\partial r \partial z}\right)_{ijk} \approx \sum_{l=1}^{N} A_{il}^{(1)} \sum_{n=1}^{P} C_{kn}^{(1)} f_{ljn}, \ \left.\frac{\partial^2 f}{\partial \theta \partial z}\right)_{ijk} \approx \sum_{m=1}^{M} B_{jm}^{(1)} \sum_{n=1}^{P} C_{kn}^{(1)} f_{imn},
\end{aligned} \tag{17b}$$

where $A^{(i)}$, $B^{(i)}$ and $C^{(i)}$ denote the weighting coefficients of the ith order derivatives of the function $f(r, \theta, z)$ with respect to the r, θ and z-directions, respectively; N, M and P are the number of grid points chosen in the r, θ and z-directions, respectively.

Governing equations of the piezoelectric and the host shells are expressed in the Appendix, that is, in Eqs. (A1)–(A4) and Eqs. (A5)–(A7), respectively. It is to be noted that $I^{(i)}$, $J^{(i)}$ and $K^{(i)}$ denote the weighting coefficients of the ith order derivatives with respect to the r, θ and z-coordinates of the shell, respectively. The steady-state temperature fields of the host shell and the piezoelectric layers are presented in the Appendix, that is, in Eqs. (A8) and (A9), respectively. Boundary conditions of the piezoelectric and the shell layers as defined in Eqs. (12a) and (12b) are expressed as:

Piezoelectric layer:

$$\begin{aligned}
&c_{11}\left\{\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} + \frac{1}{2}\left(\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk}\right)^2\right\} + c_{12}\left\{\frac{1}{r_i} u_{ijk} + \frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} v_{imk} + \frac{1}{2}\left(\frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} u_{imk}\right)^2\right\}\\
&\quad + c_{13}\left\{\sum_{n=1}^{P} C_{kn}^{(1)} w_{ijn} + \frac{1}{2}\left(\sum_{n=1}^{P} C_{kn}^{(1)} u_{ijn}\right)^2\right\} + e_{11}\sum_{l=1}^{N} A_{il}^{(1)} \psi_{ljk} - \beta_r T_{ijk} + P(r, \theta, z) = 0,
\end{aligned} \tag{18a}$$

$$c_{55}\left\{\sum_{n=1}^{P} C_{kn}^{(1)} u_{ijn} + \sum_{l=1}^{N} A_{il}^{(1)} w_{ljk} + \sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} \sum_{n=1}^{P} C_{kn}^{(1)} u_{ijn}\right\} + e_{53}\sum_{n=1}^{P} C_{kn}^{(1)} \psi_{ijn} = 0, \tag{18b}$$

$$\begin{aligned}
&c_{66}\left\{\frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} u_{imk} + \sum_{l=1}^{N} A_{il}^{(1)} v_{ljk} - \frac{1}{r_i} v_{ijk} + \frac{1}{r_i}\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} \sum_{m=1}^{M} B_{jm}^{(1)} u_{imk}\right\}\\
&\quad + \frac{e_{62}}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} \psi_{imk} = 0,
\end{aligned} \tag{18c}$$

$$\begin{aligned}
&e_{11}\left\{\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} + \frac{1}{2}\left(\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk}\right)^2\right\} + e_{21}\left\{\frac{1}{r_i} u_{ijk} + \frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} v_{imk}\right.\\
&\quad \left. + \frac{1}{2}\left(\frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} u_{imk}\right)^2\right\} + e_{31}\left\{\sum_{n=1}^{P} C_{kn}^{(1)} w_{ijn} + \frac{1}{2}\left(\sum_{n=1}^{P} C_{kn}^{(1)} u_{ijn}\right)^2\right\} - \eta_{11}\sum_{l=1}^{N} A_{il}^{(1)} \psi_{ljk} + p_r T_{ijk} = 0.
\end{aligned} \tag{18d}$$

Shell layer:

$$\frac{E}{(1+\upsilon)(1-2\upsilon)}\left\{(1-\upsilon)\left\{\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}+\frac{1}{2}\left(\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\right)^{2}\right\}\right.$$
$$+\upsilon\left\{\frac{1}{r_i}u_{ijk}+\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} v_{imk}+\frac{1}{2r_i^2}\left(\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\right)^{2}\right\}+\upsilon\left\{\sum_{n=1}^{P} K_{kn}^{(1)} w_{ijn}\right.$$
$$\left.\left.+\frac{1}{2}\left(\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}\right)^{2}\right\}\right\}-\frac{E\alpha}{(1-2\upsilon)}T_{ijk}+P(\theta,z)=0, \tag{19a}$$

$$\frac{E}{2(1+\upsilon)}\left\{\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}+\sum_{l=1}^{N} I_{il}^{(1)} w_{ljk}+\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}\right\}=0, \tag{19b}$$

$$\frac{E}{2(1+\upsilon)}\left\{\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}+\sum_{l=1}^{N} I_{il}^{(1)} v_{ljk}-\frac{1}{r_i}v_{ijk}+\frac{1}{r_i}\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\right\}=0. \tag{19c}$$

In order to proceed with numerical computations, sampling points in the radial and axial directions are defined using the Chebyshev polynomials, while for the circumferential direction, a uniform distribution is used. Coordinates of grid points used in the r, θ and z-directions are defined as follows:

$$\begin{aligned} &r_{k_1}=a+\tfrac{b-a}{2}\left\{1-\cos\left(\tfrac{k_1-1}{N-1}\pi\right)\right\},\quad k_1=1,\ldots,N;\quad \theta_{k_2}=2\pi\left(\tfrac{k_2-1}{M}\right),\quad k_2=1,\ldots,M;\\ &z_{k_3}=\tfrac{L}{2}\left\{1-\cos\left(\tfrac{k_3-1}{P-1}\pi\right)\right\},\quad k_3=1,\ldots,P. \end{aligned} \tag{20}$$

5 Numerical results and discussions

In this section, two types of shells are studied.

Shell 1: A two-layered shell made of one layer, Monel/Zirconia FGM, with one layer, PZT-4, perfectly bonded to the outer surface.

Shell 2: A three-layered shell made of one layer, Monel/Zirconia FGM, with two layers, Ba2NaNb5O15 and PZT-4, perfectly bonded to the inner and outer surfaces.

A sectional view of the cylindrical shell of the problem and the polar axes (r,θ) are shown schematically in Fig. 1.

For the spatial distribution of applied loads, the following cases are considered:

$$\text{Case } I: P(\theta)=\begin{cases}0 & 0\le\theta<\pi/4,\, 3\pi/4<\theta\le 2\pi\\ P_0 & \pi/4\le\theta\le 3\pi/4\end{cases};\quad \text{Case} II:\ P(\theta)=P_0\cos(\theta). \tag{21}$$

At first, the effectiveness of the present method is studied and the results are compared with the results reported for a long hollow cylinder made of functionally graded material [8]. It is assumed that the cylinder is made of isotropic materials graded along the r-coordinate, that is, the outer surface is made of Ni and the inner surface of TiC. Properties of these materials are listed in Table 1.

It is assumed that the variation of the modulus of elasticity E, the coefficient of thermal expansion α and the thermal conduction coefficient K obey the following power law formulation:

$$\begin{aligned} &E(r)=E_i+(E_o-E_i)\left(\frac{r-a}{b-a}\right)^{\beta_1},\quad \alpha(r)=\alpha_i+(\alpha_o-\alpha_i)\left(\frac{r-a}{b-a}\right)^{\beta_2},\\ &K(r)=K_i+(K_o-K_i)\left(\frac{r-a}{b-a}\right)^{\beta_3}, \end{aligned} \tag{22}$$

where E_i, α_i, K_i and E_o, α_o, K_o are the modulus of elasticity, the coefficient of thermal expansion and the thermal conduction coefficient at the inner and outer radii of the shell, respectively, and β_i $(i=1,2,3)$ are

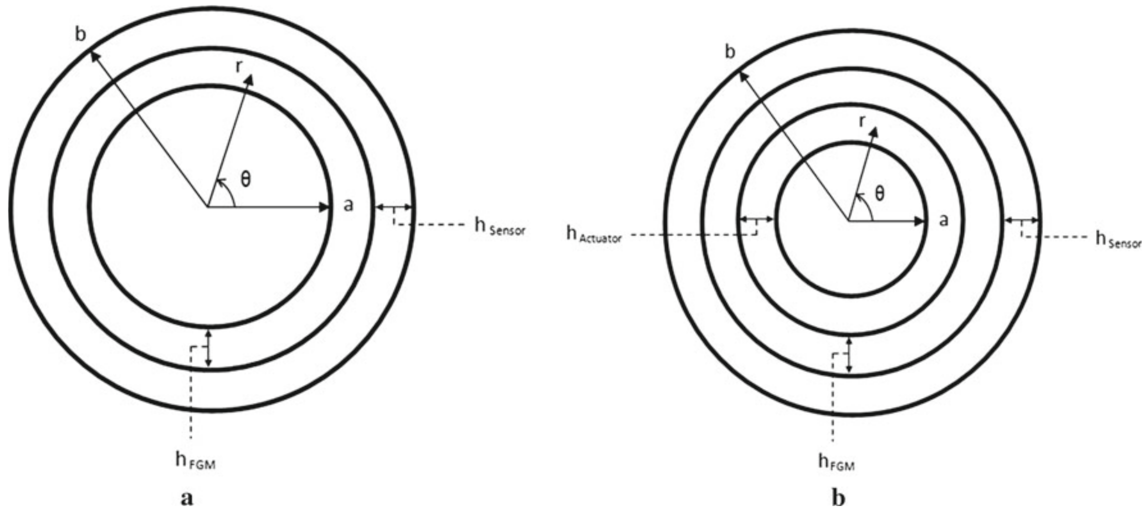

Fig. 1 A sectional view of the geometry and coordinates of the laminated shell **a** Shell 1, **b** Shell 2

Table 1 Material properties of TiC and Ni

Property	TiC	Ni
E (GPa)	460	199.5
υ	0.336	0.312
$\alpha(10^{-6}/°\text{C})$	7.4	18
K (W/m. °C)	20	90.5

their corresponding grading parameters in the radial direction of the cylindrical shell. The Poisson's ratio is assumed to be constant throughout the cylinder (i.e., $\upsilon = 0.324$). The grading index of the modulus of elasticity is chosen to be equal to 10 (i.e., $\beta_1 = 10$), and the grading indices of the coefficient of thermal expansion and the thermal conduction coefficient are assumed to be equal to 1 (i.e., $\beta_2 = \beta_3 = 1$). The thermal boundary condition is assumed to be axisymmetric, and the temperatures at the inner and the outer surfaces are taken as $T(a) = 0\,°\text{C}$ and $T(b) = 100\,°\text{C}$, respectively.

Results of the convergence study for different grid numbers are shown in Fig. 2a–c. It is observed from Fig. 2a that by increasing the number of grid points, results converge rapidly and the rate of convergence decreases as the number of grid points increases. Hence, an appropriate grid number of 20 is chosen. The radial distribution of the dimensionless thermal stresses calculated by the present method is compared with the results reported in [8] indicating a very good agreement, as shown in Fig. 2a–c.

Now, two types of shells, namely shell 1 and 2, are considered. Material properties of two piezoelectric materials are listed in Table 2 and those of Monel and Zirconia are as follows:

$$\begin{aligned}&\text{Monel:} \quad E_0 = 227.24(\text{GPa}), \alpha_0 = 15 \times 10^{-6}(1/\text{K}), K_0 = 25(\text{W/mK})\\&\text{Zirconia:} \quad E_h = 125.83(\text{GPa}), \alpha_h = 10 \times 10^{-6}(1/\text{K}), K_h = 2.09(\text{W/mK}).\end{aligned}$$

The functionally graded layer is assumed to be consisting of the inner surface made of Monel and the outer one made of Zirconia. The value of the material properties for the shell made of functionally graded material is assumed to vary according to the power law as defined in Eq. (3), and the material property parameters, m_1, m_2 and m_3, for shell 1 and 2 are as follows:

$$\text{Shell 1:} \quad m_1 = \frac{\ln(E_h/E_0)}{\ln((a + h_{\text{FGM}})/a)}, \quad m_2 = \frac{\ln(\alpha_h/\alpha_0)}{\ln((a + h_{\text{FGM}})/a)}, \quad m_3 = \frac{\ln(K_h/K_0)}{\ln((a + h_{\text{FGM}})/a)}, \tag{23a}$$

$$\begin{aligned}\text{Shell 2:} \quad m_1 &= \frac{\ln(E_h/E_0)}{\ln((a + h_{\text{Actuator}} + h_{\text{FGM}})/(a + h_{\text{Actuator}}))}, \quad m_2 = \frac{\ln(\alpha_h/\alpha_0)}{\ln((a + h_{\text{Actuator}} + h_{\text{FGM}})/(a + h_{\text{Actuator}}))},\\ m_3 &= \frac{\ln(K_h/K_0)}{\ln((a + h_{\text{Actuator}} + h_{\text{FGM}})/(a + h_{\text{Actuator}}))}.\end{aligned} \tag{23b}$$

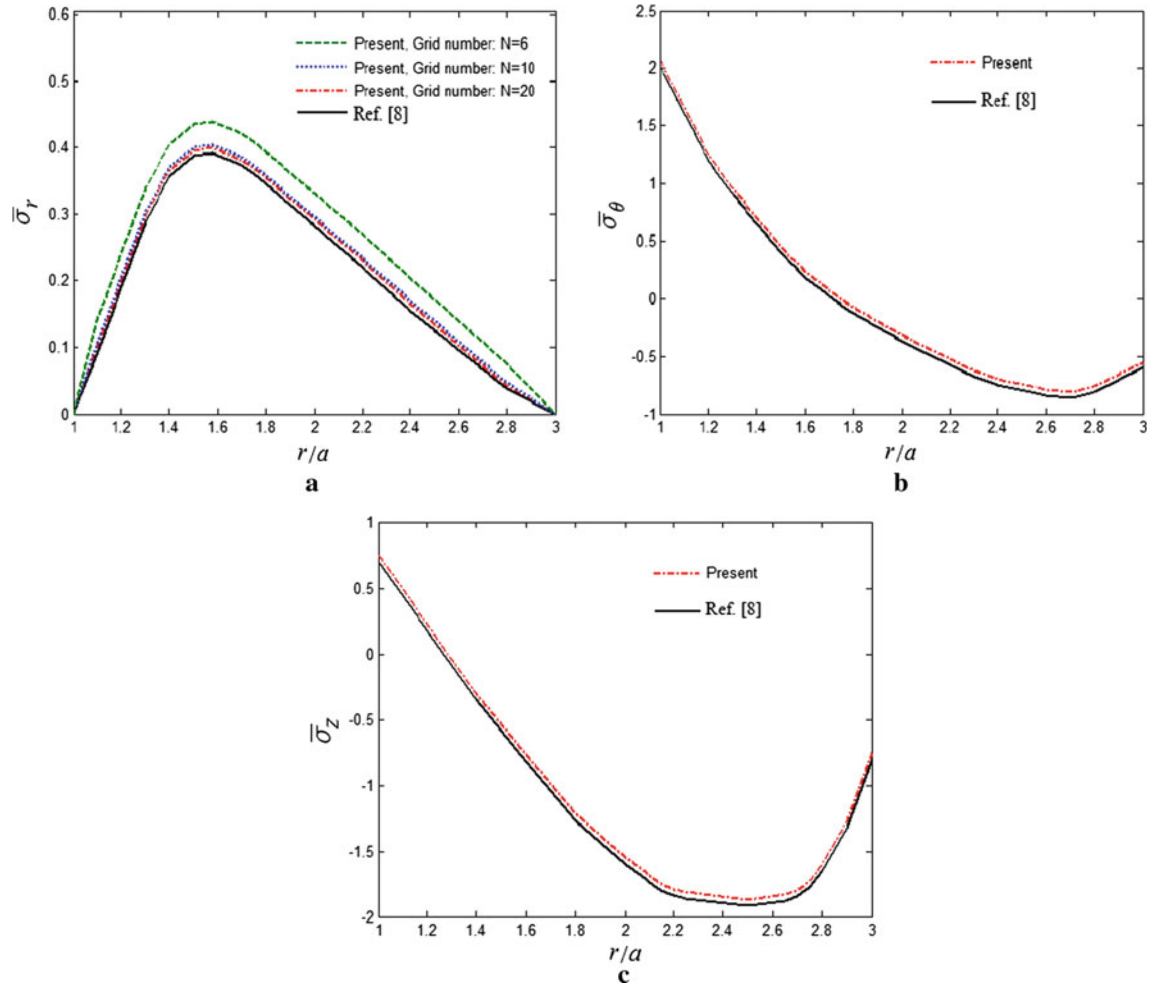

Fig. 2 Through-the-thickness variation of **a** the radial stress, **b** the circumferential stress, **c** the axial stress of an FGM cylindrical shell

Table 2 Material properties of piezoelectrics

Material	Elastic constants, GPa								
	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
PZT-4	139	78	74	139	74	115	25.6	25.6	30.5
Ba2NaNb5O15	239	104	5	247	52	135	65	66	76

	Piezoelectric constants, C/m^2					Permittivity, $10^{-9}\ C^2/Nm^2$			Pyroelectric constants, $10^{-5}\ C/m^2\ K$		
	e_{11}	e_{12}	e_{13}	e_{22}	e_{23}	η_{11}	η_{22}	η_{33}	P_1	P_2	P_3
PZT-4	−5.2	5.2	15.1	12.7	12.7	6.5	6.5	5.6	5.4	5.4	5.4
Ba2NaNb5O15	−0.4	−0.3	4.3	3.4	2.8	1.96	2.01	0.28	5.4	5.4	5.4

	Coefficient of thermal expansion, $10^{-6} 1/K$			Thermal conductivity, w/mK		
	α_r	α_θ	α_z	K_r	K_θ	K_z
PZT-4	2.62	1.97	1.97	1.5	2.1	2.1
Ba2NaNb5O15	2.458	4.396	4.396	13.9	8.6	8.6

It should be noted that all numerical results are calculated for a point located at the middle of the cylinder, that is, $z = L/2$. To better understand the problem and visualize the influence of critical parameters, dimensionless parameters of stress, displacement, temperature and electric fields are introduced:

$$(\bar{\sigma}_r, \bar{\sigma}_\theta, \bar{\sigma}_z, \bar{\sigma}_{\theta z}, \bar{\sigma}_{rz}, \bar{\sigma}_{r\theta}) = \frac{1}{P_0}(\sigma_r, \sigma_\theta, \sigma_z, \sigma_{\theta z}, \sigma_{rz}, \sigma_{r\theta}), (\bar{u}, \bar{v}, \bar{w}) = \frac{1}{b}(u, v, w), \bar{T} = \frac{T}{T_a}, \bar{\psi} = \frac{\eta_{11}\alpha_r}{p_{11}b}\psi. \tag{24}$$

A comparison is made between the results obtained for the shell type 2, using the linear and nonlinear strain–displacement relations. The effect of the nonlinear terms on results is investigated and presented in Fig. 3a–e. The temperatures at the inner and outer surfaces are assumed to be $T(a) = 100\,°\mathrm{C}$ and $T(b) = 0\,°\mathrm{C}$, respectively. The shell is subjected to the mechanical loading case II according to Eq. (21), that is, $\sigma_r = P(\theta) = 10^6 \cos(\theta)$ Pa. Comparison of the results reveals that the nonlinear terms have a significant effect on the dimensionless radial stress only. It is shown in Fig. 3a, d that values of dimensionless radial and shear stresses obtained by the nonlinear theory are higher than those of the linear theory. However, the trend is reversed for the circumferential and axial stresses, as shown in Fig. 3b, c. The dimensionless radial displacement obtained by the nonlinear theory is found to be higher than that of the linear theory, as shown in Fig. 3e.

Effects of the grading index on the distribution through the thickness of dimensionless stresses, displacement, electric and temperature fields for the shell type 1 under the loading condition I are presented in Fig. 4a–f. The temperatures at the inner and the outer radii are assumed to be 100 and 0 °C, respectively. The boundary conditions and continuity of surface tractions, radial displacement and temperature distribution across interfaces of distinct materials are well satisfied. It is revealed from Fig. 4a–e that the dimensionless radial and circumferential stresses and the dimensionless radial displacement of any point in an FG shell are lower in magnitude than those of the corresponding point in a homogeneous shell; however, the behavior for the dimensionless temperature distribution is reversed. It is found from Fig. 4b that unlike the radial and shear stresses, curves of the circumferential stress are discontinuous at the host and sensor layers interfaces. The value of the dimensionless temperature for the homogeneous shell is found to be less in comparison with that of an FG shell, as shown in Fig. 4e. It is also observed from Fig. 4e that the distribution of the temperature across the thickness in the FGM shell is nonlinear, which is due to the variation of the thermal conductivity. The dimensionless electric potential in the host shell is zero, as shown in Fig. 4f. It is found that its absolute value in the sensor layer with the FG shell is less than that of the sensor layer with the homogeneous shell. In fact, an electric potential is introduced in the sensor layer when subjected to a mechanical excitation, indicating the direct piezoelectric effect.

Next, the effect of the temperature difference at the two surfaces of the shell type 1 on stresses, radial displacement and electric potential is demonstrated in Fig. 5a–e. For the sake of simplicity, numerical calculations are carried out with the temperature of the inner surface varying as a parameter and the temperature at the outer surface of the shell being a constant, that is, $T(b) = 0\,°\mathrm{C}$. Variation of the dimensionless radial stress is displayed in Fig. 5a showing that as the temperature difference increases, the absolute values of dimensionless radial stresses decrease till $r/b \approx 0.85$ and increase afterward. The absolute value of the circumferential stress at the inner and outer boundaries increases as the temperature difference increases, as shown in Fig. 5b. It is also observed that the response curves of the dimensionless circumferential stress intersect at a radial position of about $r/b \approx 0.97$, indicating an invariant value of the circumferential stress for different values of temperature differences. Furthermore, it is revealed from Fig. 5c–e that the absolute values of the dimensionless shear stress, radial displacement and electric potential increase as the temperature difference increases. It is found that the values of the dimensionless stresses, displacement and electric fields are sensitive to the applied thermal loading. The dimensionless radial and circumferential stresses are important design parameters, since the yielding and cracking of a cylindrical shell may occur when these stresses reach some critical stress levels. As a consequence, one can choose an appropriate grading parameter to promote the safety and structural integrity of an FG shell.

Effects of the end boundary conditions on the radial distribution of dimensionless stresses, radial displacement and electric potential for the shell type 1 under the load condition I, with and without the sensor layer, are shown in Fig. 6a–e. The temperatures of the shell at the inner and outer radii are $T(a) = 100\,°\mathrm{C}$ and $T(b) = 0\,°\mathrm{C}$, respectively. Effects of presence of the sensor layer and the dependence of parameters to the edge conditions are depicted in Fig. 6a–e. It is observed from Fig. 6a, c and d that the interface continuity conditions and boundary conditions for the radial and shear stresses as well as the radial displacement are well satisfied. It is shown in Fig. 6a that the dimensionless radial stress for a clamped shell is slightly higher than that of a simply supported shell. It is also shown that when the sensor layer is used, the values of the

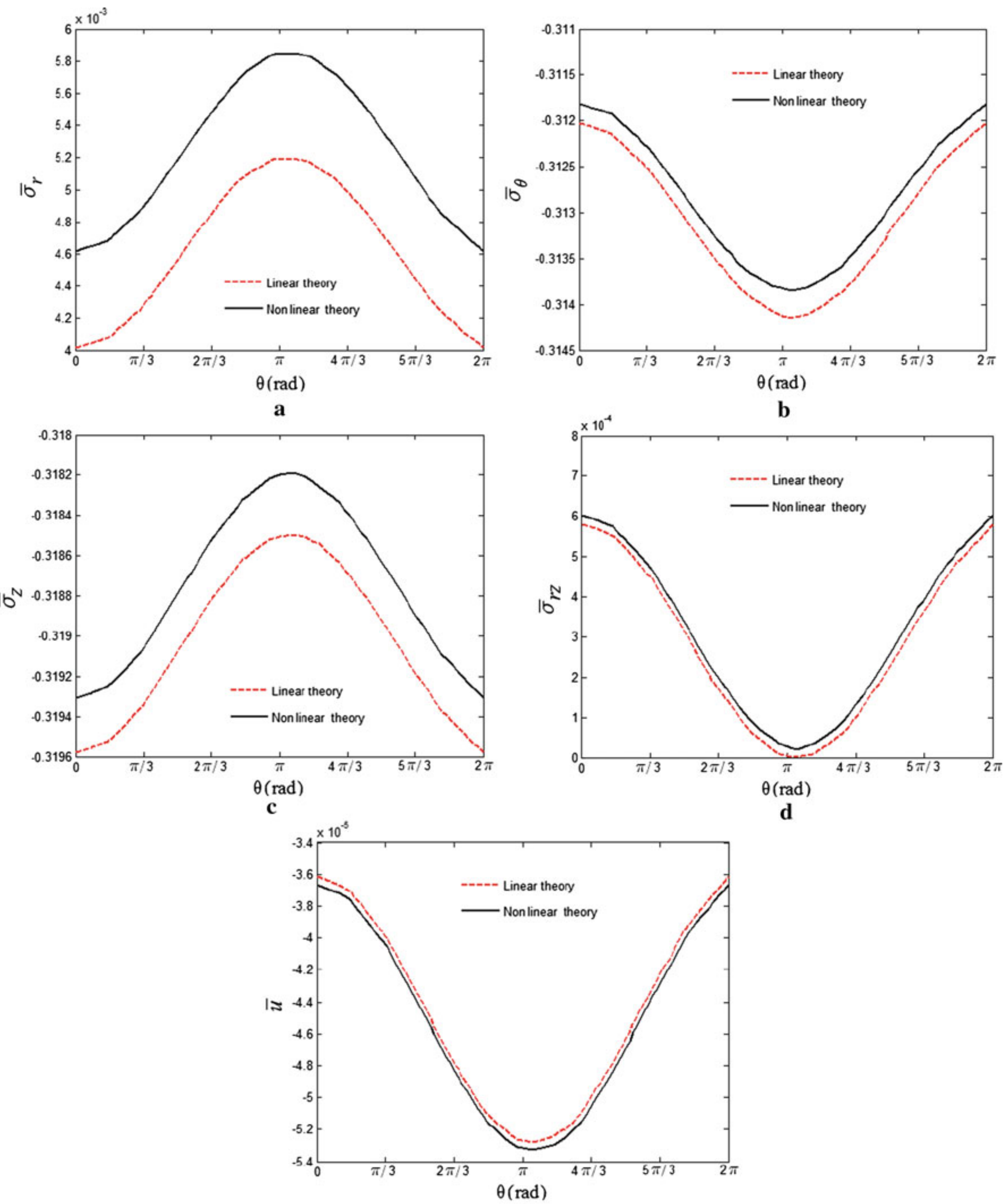

Fig. 3 Comparison of the linear and nonlinear theories on the dimensionless **a** radial stress, **b** circumferential stress, **c** axial stress, **d** shear stress, **e** radial displacement

dimensionless stresses in the host layer are reduced. However, it is noted that such a reduction for a simply supported shell is higher than that of a clamped shell. A similar trend is observed in the distribution of the shear stress, as shown in Fig. 6c. Hence, it is concluded that by using the sensor layer, the host FG shell will experience less circumferential and shear stresses, especially in case of a simply supported shell. Through-the-thickness distributions of the radial displacement with and without the sensor layer for clamped and simply supported conditions are plotted in Fig. 6d. It is observed that presence of the sensor layer reduces the value of the dimensionless radial displacement. Such reduction for a simply supported shell is higher than that of a

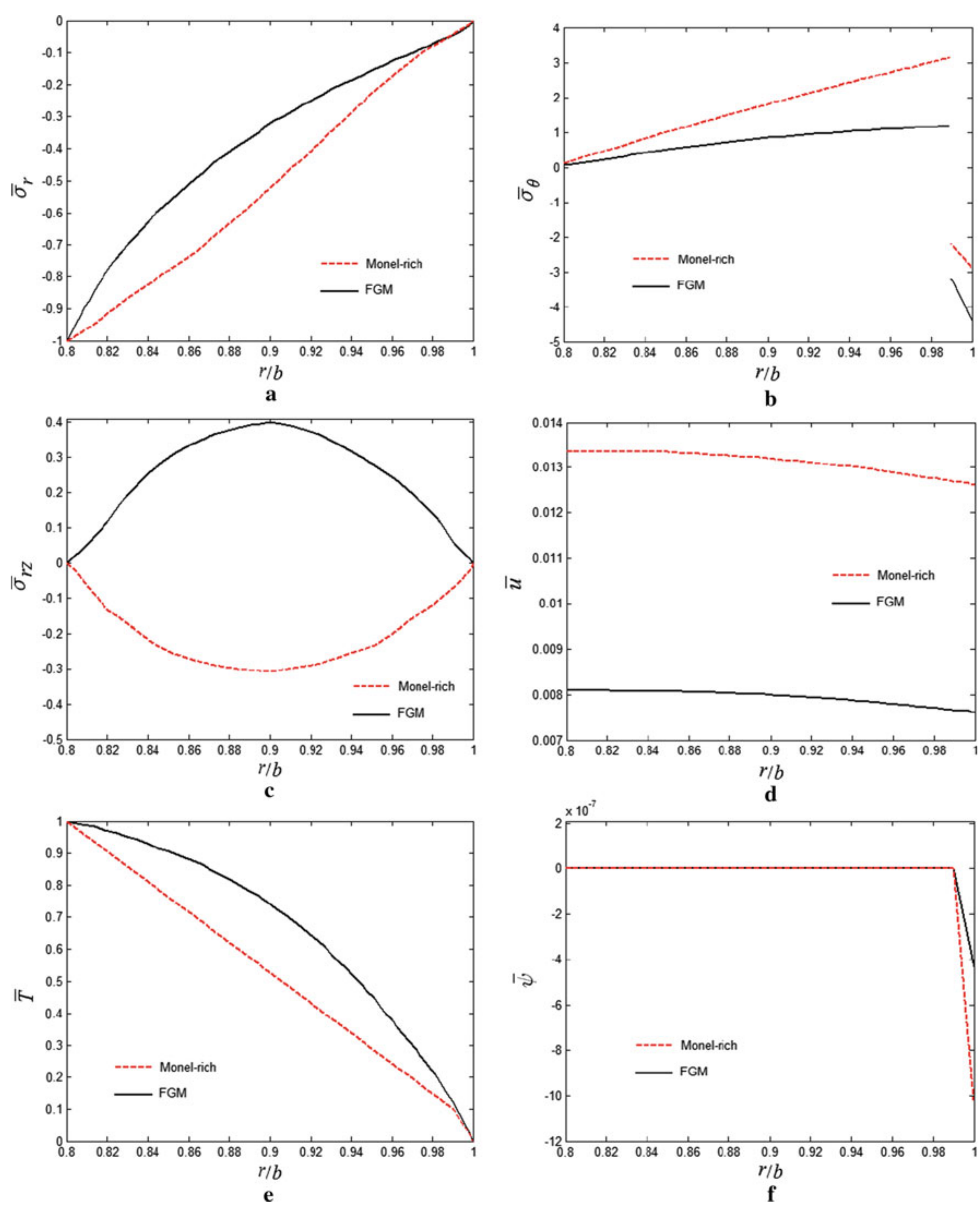

Fig. 4 Effects of the grading parameter on the dimensionless **a** radial stress, **b** circumferential stress, **c** shear stress, **d** radial displacement, **e** temperature distribution, **f** electric potential

clamped shell. The distribution of the radial displacement in the clamped condition is found to be nonlinear, while it is linear for a simply supported shell. The effect of the boundary conditions is found to be higher on the through-the-thickness distribution of the electric potential than that of other parameters, as shown in Fig. 6e. It is also noted that the dimensionless sensed voltage via the sensor layer for the simply supported shell is higher in magnitude than that of the clamped shell.

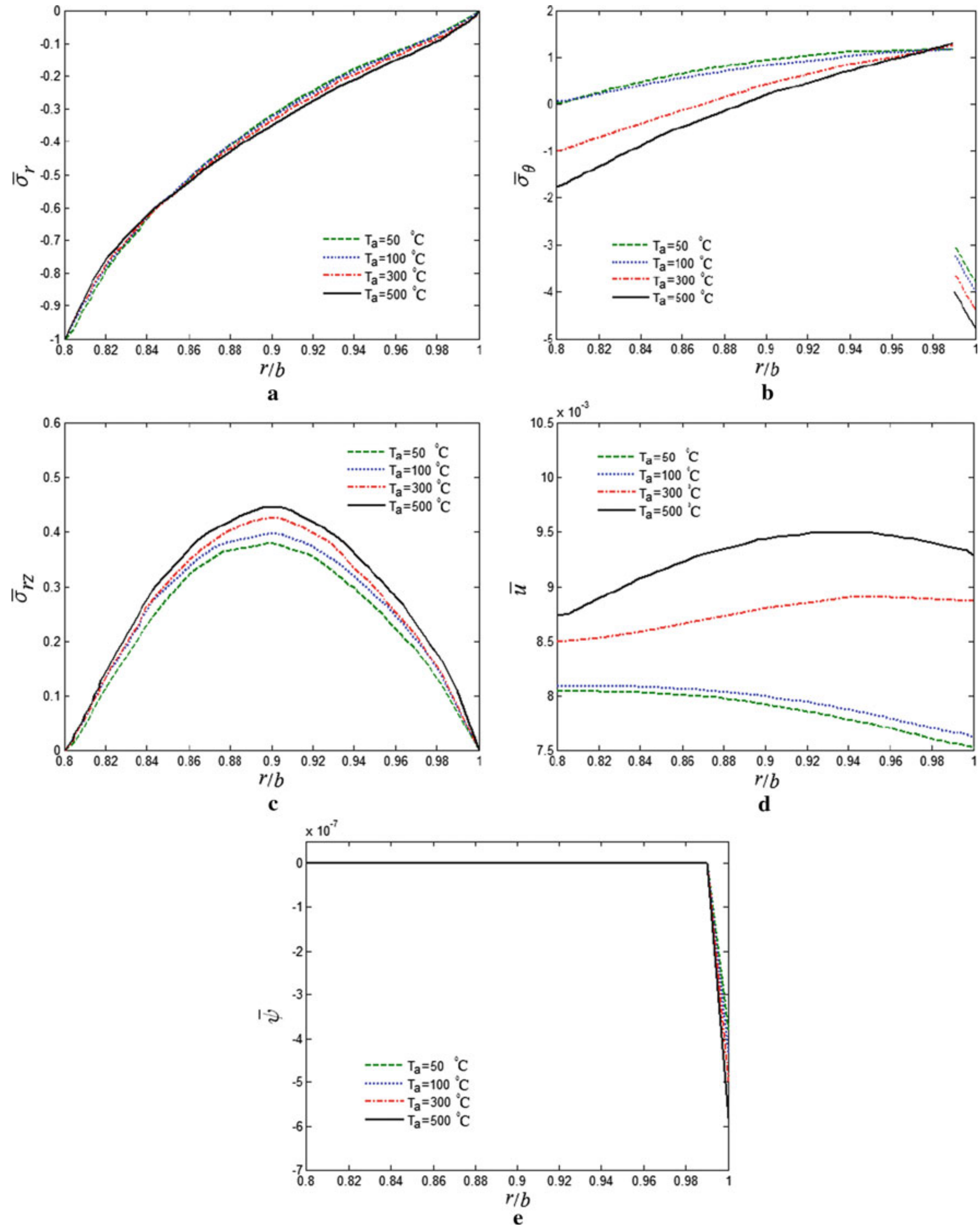

Fig. 5 Effects of temperature difference on the dimensionless **a** radial stress, **b** circumferential stress, **c** shear stress, **d** radial displacement, **e** electric potential

The effects of the ratio of the radius of the mid-plane to the thickness (i.e., S) of the shell type 1 under the load case I are shown in Fig. 7a–f. Thermal boundary conditions at the inner and outer surfaces are taken as $T(a) = 100\,°\mathrm{C}$ and $T(b) = 0\,°\mathrm{C}$, respectively. It is observed from Fig. 7a, b that the dimensionless radial and circumferential stresses increase as the parameter S increases; however, the behavior of the shear stress and radial displacement is different. The distribution of the dimensionless temperature is presented in Fig. 7e,

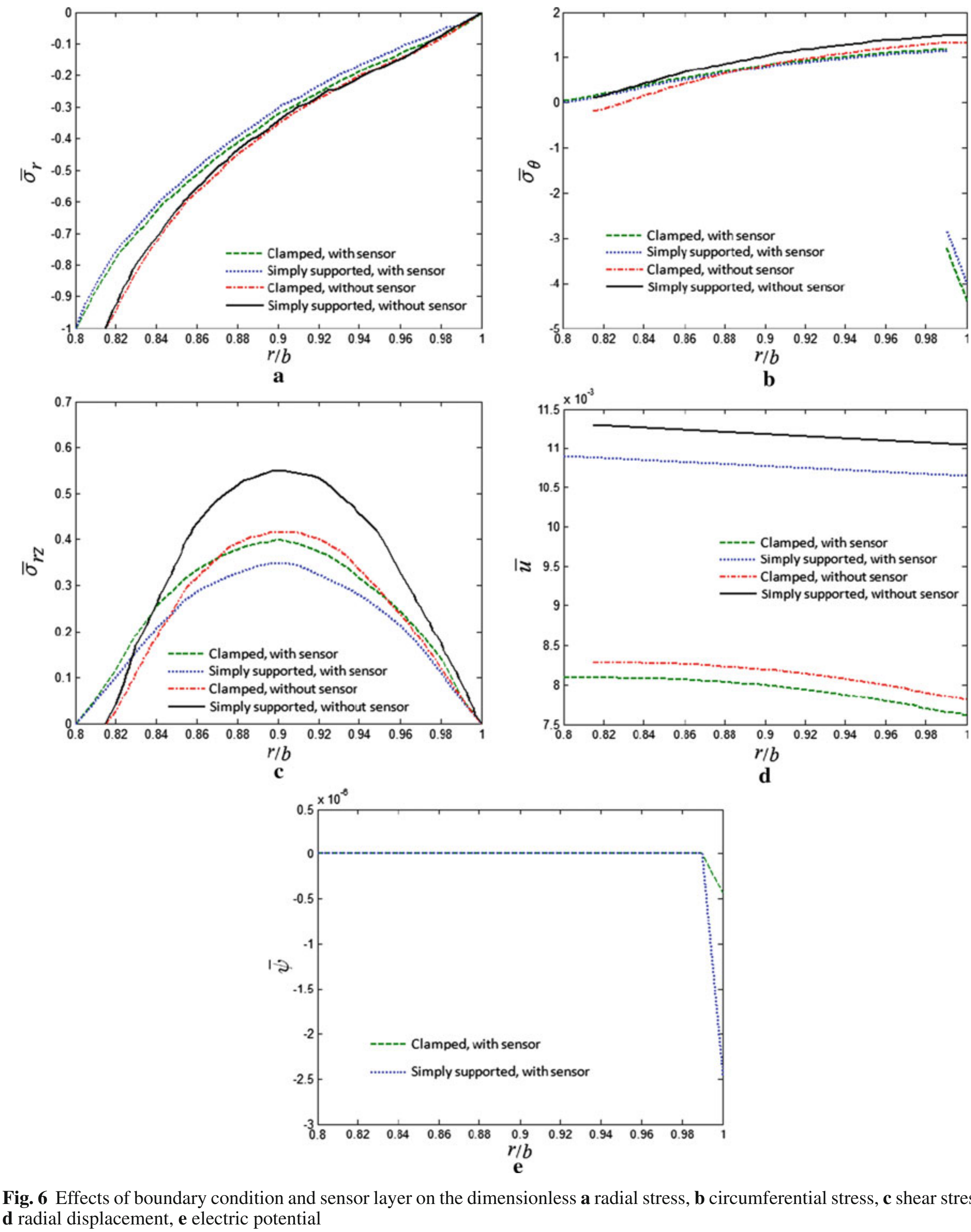

Fig. 6 Effects of boundary condition and sensor layer on the dimensionless **a** radial stress, **b** circumferential stress, **c** shear stress, **d** radial displacement, **e** electric potential

showing that the temperature increases as the parameter S increases. Variation of the electric potential to S is plotted in Fig. 7f.

Effects of the thickness ratio of the actuator to the sensor layers (h_A/h_S) for the shell type 2 under the load case II at the mid-plane of the shell are depicted in Fig. 8a–e. The stress and displacement responses of the FG host layer are found to be highly dependent to the thickness ratio. This fact is a key parameter in appropriate designing of coupled piezoelectric systems. It is observed from Fig. 8a that the maximum value of

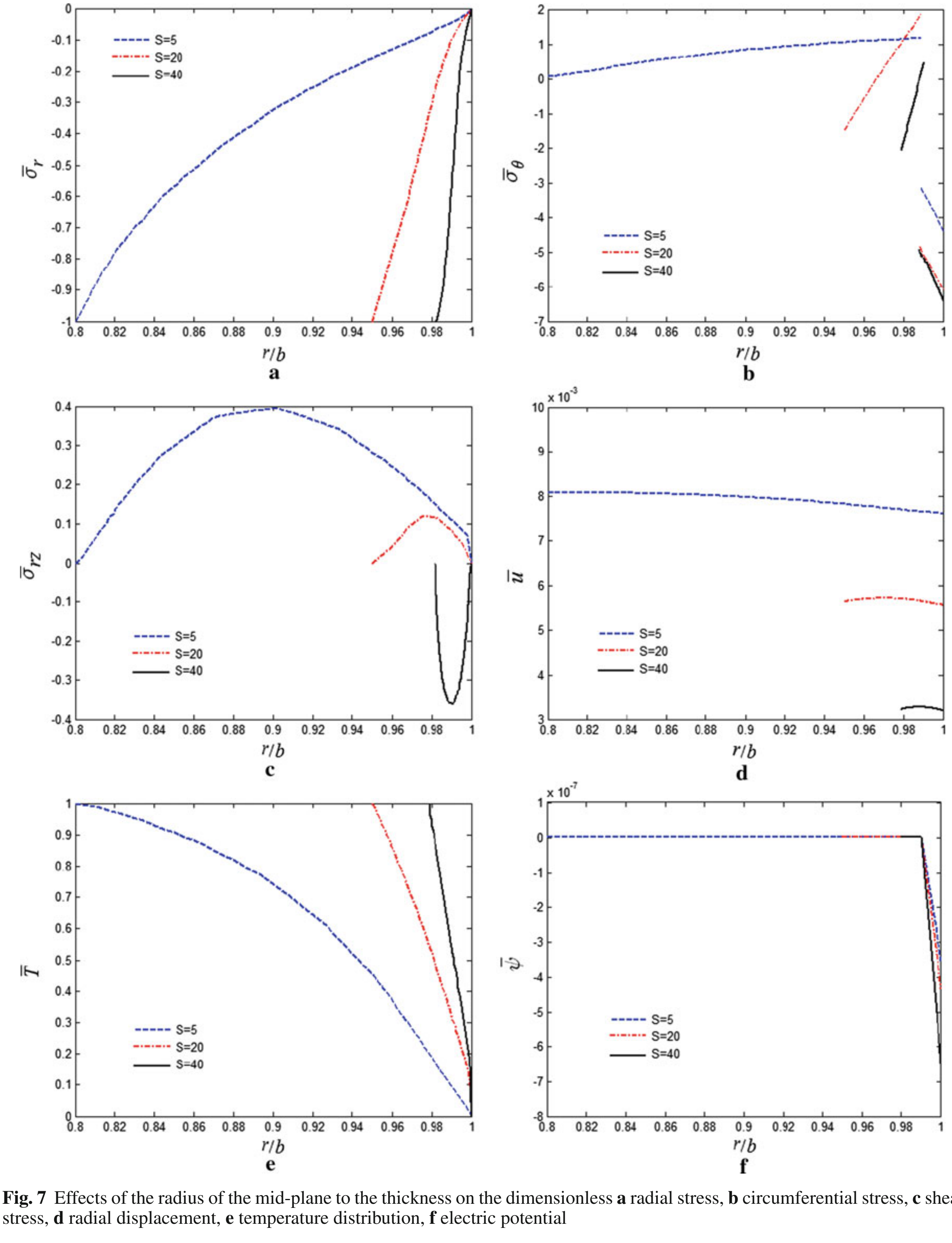

Fig. 7 Effects of the radius of the mid-plane to the thickness on the dimensionless **a** radial stress, **b** circumferential stress, **c** shear stress, **d** radial displacement, **e** temperature distribution, **f** electric potential

dimensionless radial stress occurs in case of equal thicknesses of actuator and sensor layers. It is revealed from the figures that the curves of dimensionless stresses and radial displacement shift considerably with variation of h_A/h_S.

Next, the effects of the electric excitation and ratio of the length to total thickness of the shell (L/H) are investigated in order to optimize stress and displacement responses of the cylindrical shell. The loading case

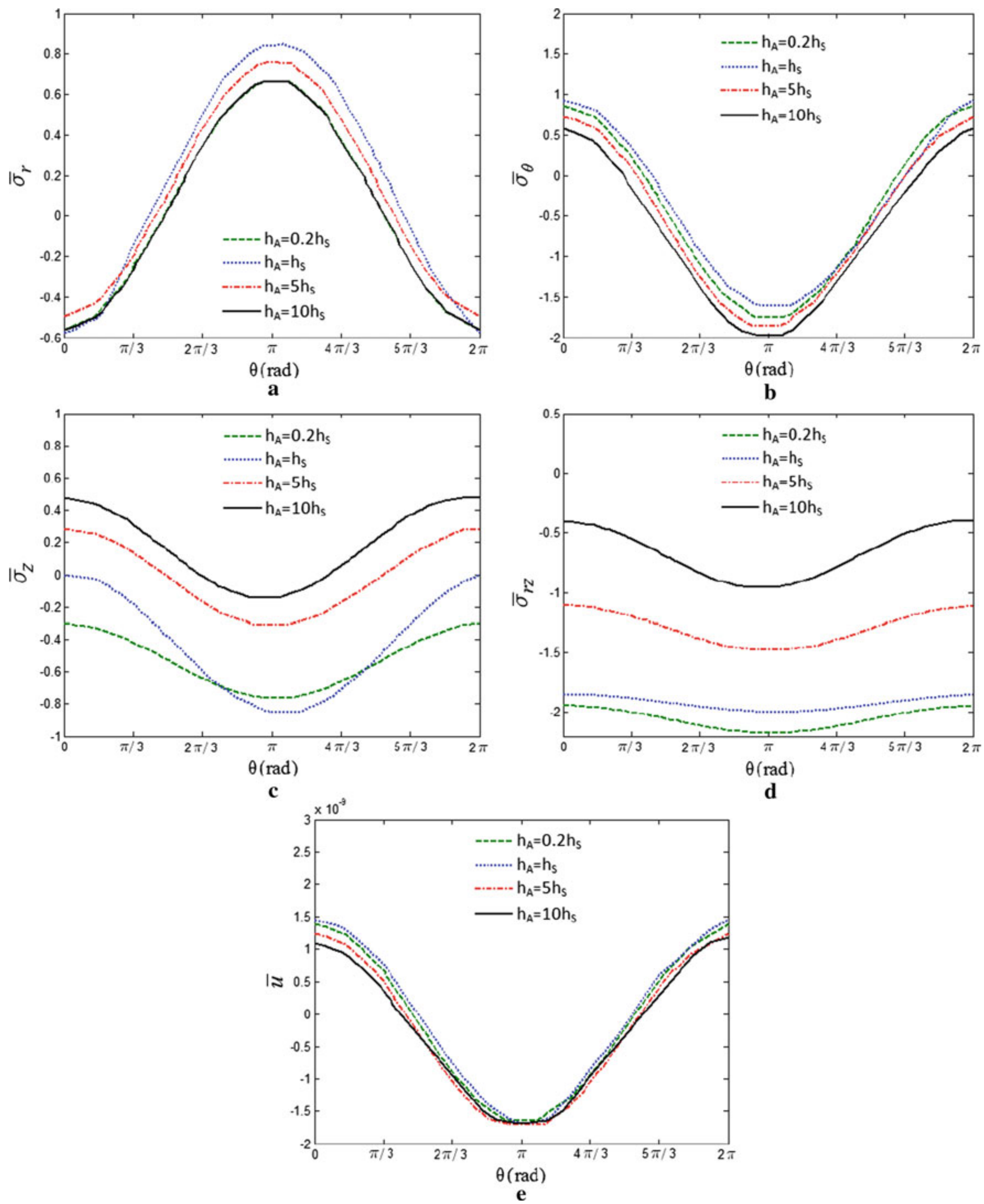

Fig. 8 Effects of the actuator's thickness to sensor's thickness on the dimensionless **a** radial stress, **b** circumferential stress, **c** axial stress, **d** shear stress, **e** radial displacement

II is chosen and the electric excitation is assumed to vary as a parameter, that is, $\psi = 100$ and 500 V. Results of stresses and displacement parameters are shown in Fig. 9a–e. It is realized from these figures that absolute values of the dimensionless stresses and radial displacement increase as the electric excitation increases and shift as L/H is changed. It is also found that for higher values of L/H, effects of electric excitation on circumferential stress and radial displacement are reduced.

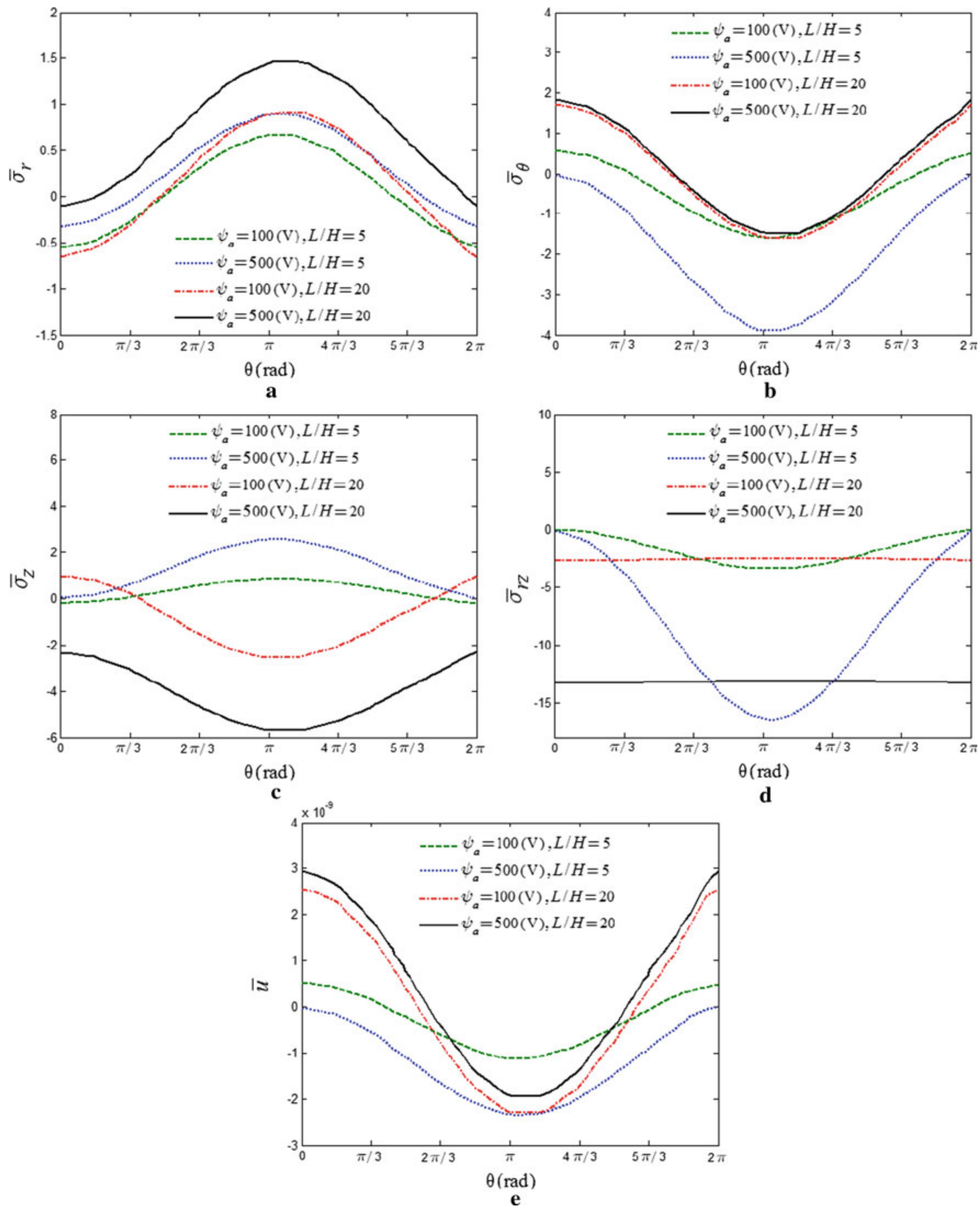

Fig. 9 Effects of electric excitation and ratio of length to total thickness on the dimensionless **a** radial stress, **b** circumferential stress, **c** axial stress, **d** shear stress, **e** radial displacement

Finally, the effect of the piezoelectric layer thickness on the behavior of the FG shell type 2 is shown in Fig. 10a–e. It is observed from Fig. 10a that values of dimensionless radial stress decrease as the piezoelectric layer becomes thinner. It is also revealed from Fig. 10b, d and e that as the piezoelectric layers become thinner, the amplitudes of variation of the dimensionless circumferential and shear stresses and radial displacement increase. However, the behavior of the dimensionless axial stress and radial displacement is the opposite. It is

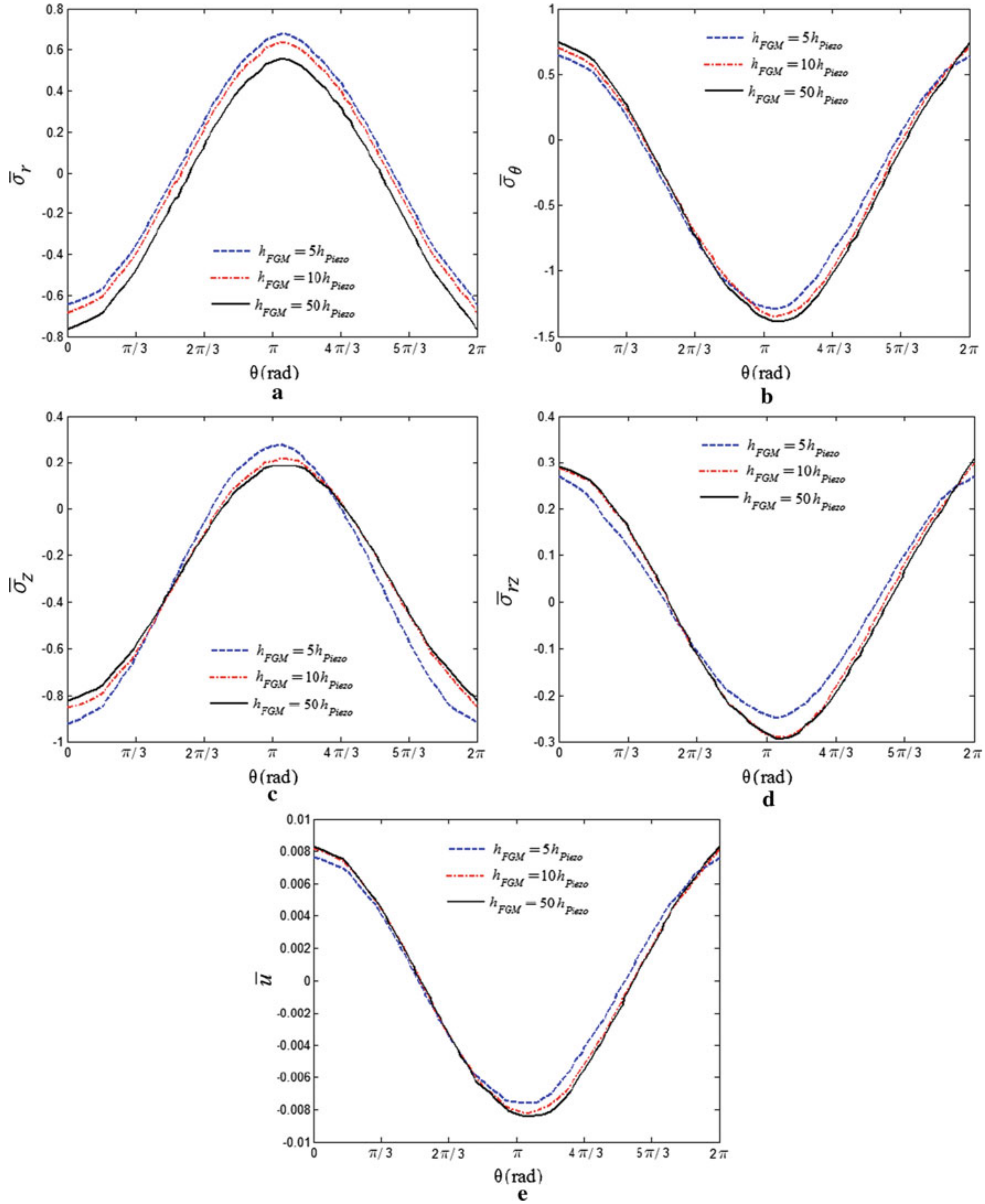

Fig. 10 Effects of piezoelectric layers thickness on the dimensionless **a** radial stress, **b** circumferential stress, **c** axial stress, **d** shear stress, **e** radial displacement

also observed that as the thickness of piezoelectric layer decreases to about $h_{\text{Piezo}} = 0.02\,h_{\text{FGM}}$, the effect of the piezoelectric layer becomes negligibly small.

6 Conclusions

A semi-analytical solution for three-dimensional nonlinear thermo-elastic problem of an FG hollow cylindrical shell bonded with piezoelectric layers and motivated by thermo-electro-mechanical loads is presented.

The governing differential equations are solved using two versions of differential quadrature, namely FDQ and PDQ. Boundary conditions at edges and the interface continuity conditions between adjacent layers are exactly satisfied. The numerical results reveal that the material inhomogeneity and presence of piezoelectric layers play important roles on the distribution of stress, displacement, electric and thermal fields of the cylindrical shell. From the present study, the following conclusions are obtained:

– In a functionally graded shell, distributions of the temperature and the radial stress through the thickness of the shell due to the power law variation of material properties across the thickness are nonlinear.
– By employing the sensor layer of appropriate dimensions and selecting a suitable boundary condition, values of the stress and displacement fields in the host shell can be optimized.
– Effects of boundary conditions on the distribution of electric potential in the shell are the highest and on the radial stress are the lowest.
– Values of stresses and displacements in an FG shell are lower than those of a corresponding point in a homogeneous shell.
– Values of the radial displacement, the transverse shear stress and the radial stress increase as the ratio of the radius of the mid-plane to the thickness increases.
– Electric excitation has intense effect on values of stresses and displacement fields, that is, values of stresses and radial displacement increase as the electric excitation increases.
– The influence of piezoelectricity becomes negligibly small, when the thickness of piezoelectric layer reduces to about 2 % of the thickness of the host shell.

Appendix

Governing equations of the piezoelectric shell are expressed as:

$$
\begin{aligned}
&c_{11}\left\{\sum_{l=1}^{N} A_{il}^{(2)} u_{ljk} + \sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} \sum_{l=1}^{N} A_{il}^{(2)} u_{ljk}\right\} + c_{12}\left\{\frac{1}{r_i}\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} - \frac{1}{r_i^2} u_{ijk} - \frac{1}{r_i^2}\sum_{m=1}^{M} B_{jm}^{(1)} v_{imk}\right.\\
&\left. +\frac{1}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M} A_{il}^{(1)} B_{jm}^{(1)} v_{lmk} - \frac{1}{r_i^3}\left(\sum_{m=1}^{M} B_{jm}^{(1)} u_{imk}\right)^2 + \frac{1}{r_i^2}\sum_{m=1}^{M} B_{jm}^{(1)} u_{imk} \sum_{l=1}^{N}\sum_{m=1}^{M} A_{il}^{(1)} B_{jm}^{(1)} u_{lmk}\right\}\\
&+c_{13}\left\{\sum_{l=1}^{N}\sum_{n=1}^{P} A_{il}^{(1)} C_{kn}^{(1)} w_{ljn} + \sum_{n=1}^{P} C_{kn}^{(1)} u_{ijn} \sum_{l=1}^{N}\sum_{n=1}^{P} A_{il}^{(1)} C_{kn}^{(1)} u_{ljn}\right\} + e_{11}\sum_{l=1}^{N} A_{il}^{(2)} \psi_{ljk}\\
&-\beta_r \sum_{l=1}^{N} A_{il}^{(1)} T_{ljk} + \frac{1}{r_i}\left\{c_{11}\left\{\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} + \frac{1}{2}\left(\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk}\right)^2\right\} + c_{12}\left\{\frac{1}{r_i} u_{ijk} + \frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} v_{imk}\right.\right.\\
&\left. +\frac{1}{2r_i^2}\left(\sum_{m=1}^{M} B_{jm}^{(1)} u_{imk}\right)^2\right\} + c_{13}\left\{\sum_{n=1}^{P} C_{kn}^{(1)} w_{ijn} + \frac{1}{2}\left(\sum_{n=1}^{P} C_{kn}^{(1)} u_{ijn}\right)^2\right\} + e_{11}\sum_{l=1}^{N} A_{il}^{(1)} \psi_{ljk} - \beta_r T_{ijk}\\
&-c_{12}\left\{\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} + \frac{1}{2}\left(\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk}\right)^2\right\} - c_{22}\left\{\frac{1}{r_i} u_{ijk} + \frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} v_{imk} + \frac{1}{2r_i^2}\left(\sum_{m=1}^{M} B_{jm}^{(1)} u_{imk}\right)^2\right\}\\
&\left.-c_{23}\left\{\sum_{n=1}^{P} C_{kn}^{(1)} w_{ijn} + \frac{1}{2}\left(\sum_{n=1}^{P} C_{kn}^{(1)} u_{ijn}\right)^2\right\} - e_{21}\sum_{l=1}^{N} A_{il}^{(1)} \psi_{ljk} + \beta_\theta T_{ijk}\right\} + \frac{1}{r_i}\left\{c_{66}\left\{\frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(2)} v_{imk}\right.\right.\\
&+\sum_{l=1}^{N}\sum_{m=1}^{M} A_{il}^{(1)} B_{jm}^{(1)} v_{lmk} - \frac{1}{r_i}\sum_{m=1}^{M} B_{jm}^{(1)} v_{imk} + \frac{1}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M} A_{il}^{(1)} B_{jm}^{(1)} u_{lmk} \sum_{m=1}^{M} B_{jm}^{(1)} u_{imk}\\
&\left.\left.+\frac{1}{r_i}\sum_{l=1}^{N} A_{il}^{(1)} u_{ljk} \sum_{m=1}^{M} B_{jm}^{(2)} u_{imk}\right\} + \frac{e_{62}}{r_i}\sum_{m=1}^{M} B_{jm}^{(2)} \psi_{imk}\right\} + c_{55}\left\{\sum_{n=1}^{P} C_{kn}^{(2)} u_{ijn}\right.
\end{aligned}
$$

$$+\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}w_{ljn}+\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}u_{ljn}+\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{n=1}^{P}C_{kn}^{(2)}u_{ijn}\Bigg\}$$
$$+e_{53}\sum_{n=1}^{P}C_{kn}^{(2)}\psi_{ijn}=0, \tag{A1}$$

$$\begin{aligned}&\frac{1}{r_i}\left\{c_{12}\left\{\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}u_{lmk}+\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}u_{lmk}\right\}+c_{22}\left\{\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\right.\right.\\
&\left.+\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(2)}v_{imk}+\frac{1}{r_i^2}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\sum_{m=1}^{M}B_{jm}^{(2)}u_{imk}\right\}+c_{23}\left\{\sum_{m=1}^{M}\sum_{n=1}^{P}B_{jm}^{(1)}C_{kn}^{(1)}w_{imn}\right.\\
&\left.\left.+\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\sum_{m=1}^{M}\sum_{n=1}^{P}B_{jm}^{(1)}C_{kn}^{(1)}u_{imn}\right\}+e_{21}\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}\psi_{lmk}\right\}+c_{44}\left\{\sum_{n=1}^{P}C_{kn}^{(2)}v_{ijn}\right.\\
&+\frac{1}{r_i}\sum_{m=1}^{M}\sum_{n=1}^{P}B_{jm}^{(1)}C_{kn}^{(1)}w_{imn}+\frac{1}{r_i}\sum_{m=1}^{M}\sum_{n=1}^{P}B_{jm}^{(1)}C_{kn}^{(1)}u_{imn}\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\\
&\left.+\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\sum_{n=1}^{P}C_{kn}^{(2)}u_{ijn}\right\}+c_{66}\left\{-\frac{1}{r_i^2}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}+\frac{1}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}u_{lmk}\right.\\
&+\sum_{l=1}^{N}A_{il}^{(2)}v_{ljk}-\frac{1}{r_i}\sum_{l=1}^{N}A_{il}^{(1)}v_{ljk}+\frac{v_{ijk}}{r_i^2}-\frac{1}{r_i^2}\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\\
&\left.+\frac{1}{r_i}\sum_{l=1}^{N}A_{il}^{(2)}u_{ljk}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}+\frac{1}{r_i}\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}u_{lmk}\right\}-\frac{e_{62}}{r_i^2}\sum_{m=1}^{M}B_{jm}^{(1)}\psi_{imk}\\
&+\frac{e_{62}}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}\psi_{lmk}+\frac{2}{r_i}\left\{c_{66}\left\{\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}+\sum_{l=1}^{N}A_{il}^{(1)}v_{ljk}\right.\right.\\
&\left.\left.-\frac{1}{r_i}v_{ijk}+\frac{1}{r_i}\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\right\}+\frac{e_{62}}{r_i}\sum_{m=1}^{M}B_{jm}^{(1)}\psi_{imk}\right\}=0,\end{aligned} \tag{A2}$$

$$\begin{aligned}&c_{13}\left\{\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}u_{ljn}+\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}u_{ljn}\right\}+c_{23}\left\{\frac{1}{r_i}\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\right.\\
&\left.+\frac{1}{r_i}\sum_{m=1}^{M}\sum_{n=1}^{P}B_{jm}^{(1)}C_{kn}^{(1)}v_{imn}+\frac{1}{r_i^2}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\sum_{m=1}^{M}\sum_{n=1}^{P}B_{jm}^{(1)}C_{kn}^{(1)}u_{imn}\right\}+c_{33}\left\{\sum_{n=1}^{P}C_{kn}^{(2)}w_{ijn}\right.\\
&\left.+\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\sum_{n=1}^{P}C_{kn}^{(2)}u_{ijn}\right\}+e_{31}\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}\psi_{ljn}-\beta_z\sum_{n=1}^{P}C_{kn}^{(1)}T_{ijn}\\
&+c_{55}\left\{\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}u_{ljn}+\sum_{l=1}^{N}A_{il}^{(2)}w_{ljk}+\sum_{l=1}^{N}A_{il}^{(2)}u_{ljk}\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\right.\\
&\left.+\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}u_{ljn}\right\}+e_{53}\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}\psi_{ljn}\\
&+\frac{c_{44}}{r_i}\left\{\sum_{m=1}^{M}\sum_{n=1}^{P}B_{jm}^{(1)}C_{kn}^{(1)}v_{imn}+\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(2)}w_{imk}+\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(2)}u_{imk}\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\right.\end{aligned}$$

$$+\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\sum_{m=1}^{M}\sum_{n=1}^{P}B_{jm}^{(1)}C_{kn}^{(1)}u_{imn}\Bigg\}+\frac{c_{55}}{r_i}\Bigg\{\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}+\sum_{l=1}^{N}A_{il}^{(1)}w_{ljk}$$
$$+\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\Bigg\}+\frac{e_{53}}{r_i}\sum_{n=1}^{P}C_{kn}^{(1)}\psi_{ijn}=0, \tag{A3}$$

$$e_{11}\Bigg\{\sum_{l=1}^{N}A_{il}^{(2)}u_{ljk}+\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{l=1}^{N}A_{il}^{(2)}u_{ljk}\Bigg\}+e_{21}\Bigg\{\frac{1}{r_i}\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}-\frac{1}{r_i^2}u_{ijk}$$
$$-\frac{1}{r_i^2}\sum_{m=1}^{M}B_{jm}^{(1)}v_{imk}+\frac{1}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}v_{lmk}-\frac{1}{r_i^3}\left(\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\right)^2$$
$$+\frac{1}{r_i^2}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}u_{lmk}\Bigg\}+e_{31}\Bigg\{\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}w_{ljn}$$
$$+\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}u_{ljn}\Bigg\}-\eta_{11}\sum_{l=1}^{N}A_{il}^{(2)}\psi_{ljk}+p_r\sum_{l=1}^{N}A_{il}^{(1)}T_{ljk}$$
$$+\frac{1}{r_i}\Bigg\{e_{11}\Bigg\{\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}+\frac{1}{2}\left(\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\right)^2\Bigg\}+e_{21}\Bigg\{\frac{1}{r_i}u_{ijk}+\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(1)}v_{imk}$$
$$+\frac{1}{2r_i^2}\left(\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}\right)^2\Bigg\}+e_{31}\Bigg\{\sum_{n=1}^{P}C_{kn}^{(1)}w_{ijn}+\frac{1}{2}\left(\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\right)^2\Bigg\}-\eta_{11}\sum_{l=1}^{N}A_{il}^{(1)}\psi_{ljk}$$
$$+p_rT_{ijk}\}+\frac{1}{r_i}\Bigg\{e_{62}\Bigg\{\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(2)}u_{imk}+\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}v_{lmk}-\frac{1}{r_i}\sum_{m=1}^{M}B_{jm}^{(1)}v_{imk}$$
$$+\frac{1}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M}A_{il}^{(1)}B_{jm}^{(1)}u_{lmk}\sum_{m=1}^{M}B_{jm}^{(1)}u_{imk}+\frac{1}{r_i}\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{m=1}^{M}B_{jm}^{(2)}u_{imk}\}$$
$$-\frac{\eta_{22}}{r_i}\sum_{m=1}^{M}B_{jm}^{(2)}\psi_{imk}\Bigg\}+e_{53}\Bigg\{\sum_{n=1}^{P}C_{kn}^{(2)}u_{ijn}+\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}w_{ljn}$$
$$+\sum_{n=1}^{P}C_{kn}^{(1)}u_{ijn}\sum_{l=1}^{N}\sum_{n=1}^{P}A_{il}^{(1)}C_{kn}^{(1)}u_{ljn}+\sum_{l=1}^{N}A_{il}^{(1)}u_{ljk}\sum_{n=1}^{P}C_{kn}^{(2)}u_{ijn}\Bigg\}$$
$$-\eta_{33}\sum_{n=1}^{P}C_{kn}^{(2)}\psi_{ijn}+p_z\sum_{n=1}^{P}C_{kn}^{(1)}T_{ijn}=0. \tag{A4}$$

Governing equations of the host shell are expressed as:

$$\frac{E}{(1+\upsilon)(1-2\upsilon)}\Bigg\{(1-\upsilon)\Bigg\{\sum_{l=1}^{N}I_{il}^{(2)}u_{ljk}+\sum_{l=1}^{N}I_{il}^{(1)}u_{ljk}\sum_{l=1}^{N}I_{il}^{(2)}u_{ljk}\Bigg\}+\upsilon\Bigg\{\frac{1}{r_i}\sum_{l=1}^{N}I_{il}^{(1)}u_{ljk}$$
$$-\frac{1}{r_i^2}u_{ijk}-\frac{1}{r_i^2}\sum_{m=1}^{M}J_{jm}^{(1)}v_{imk}+\frac{1}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M}I_{il}^{(1)}J_{jm}^{(1)}v_{lmk}-\frac{1}{r_i^3}\left(\sum_{m=1}^{M}J_{jm}^{(1)}u_{imk}\right)^2$$
$$+\frac{1}{r_i^2}\sum_{m=1}^{M}J_{jm}^{(1)}u_{imk}\sum_{l=1}^{N}\sum_{m=1}^{M}I_{il}^{(1)}J_{jm}^{(1)}u_{lmk}\Bigg\}+\upsilon\Bigg\{\sum_{l=1}^{N}\sum_{n=1}^{P}I_{il}^{(1)}K_{kn}^{(1)}w_{ljn}$$

$$+\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn} \sum_{l=1}^{N}\sum_{n=1}^{P} I_{il}^{(1)} K_{kn}^{(1)} u_{ljn}\Bigg\}\Bigg\} - \frac{E\alpha}{(1-2\upsilon)} \sum_{l=1}^{N} I_{il}^{(1)} T_{ljk} - \frac{T_{ijk}}{(1-2\upsilon)}$$

$$\times \left\{E_i \frac{m_1}{a^{m_1}} r_i^{m_1-1} + \alpha_i \frac{m_2}{a^{m_2}} r_i^{m_2-1}\right\} + \frac{E_i \frac{m_1}{a^{m_1}} r_i^{m_1-1}}{(1+\upsilon)(1-2\upsilon)} \Bigg\{(1-\upsilon)\Bigg\{\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}$$

$$+\frac{1}{2}\left(\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\right)^2\Bigg\} + \upsilon\left\{\frac{1}{r_i} u_{ijk} + \frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} v_{imk} + \frac{1}{2r_i^2}\left(\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\right)^2\right\}$$

$$+\upsilon\left\{\sum_{n=1}^{P} K_{kn}^{(1)} w_{ijn} + \frac{1}{2}\left(\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}\right)^2\right\}\Bigg\} + \frac{E}{(1+\upsilon)(1-2\upsilon)r_i}\Bigg\{(1-\upsilon)\Bigg\{\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}$$

$$+\frac{1}{2}\left(\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\right)^2\Bigg\} + \upsilon\left\{\frac{1}{r_i} u_{ijk} + \frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} v_{imk} + \frac{1}{2r_i^2}\left(\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\right)^2\right\}$$

$$+\upsilon\left\{\sum_{n=1}^{P} K_{kn}^{(1)} w_{ijn} + \frac{1}{2}\left(\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}\right)^2\right\}\Bigg\} - \frac{E}{(1+\upsilon)(1-2\upsilon)r_i}\Bigg\{(1-\upsilon)\Bigg\{\frac{1}{r_i} u_{ijk} + \frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} v_{imk}$$

$$+\frac{1}{2r_i^2}\left(\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\right)^2\Bigg\} + \upsilon\left\{\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk} + \frac{1}{2}\left(\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\right)^2\right\} + \upsilon\Bigg\{\sum_{n=1}^{P} K_{kn}^{(1)} w_{ijn}$$

$$+\frac{1}{2}\left(\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}\right)^2\Bigg\}\Bigg\} + \frac{E}{2(1+\upsilon)r_i}\Bigg\{\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(2)} u_{imk} + \sum_{l=1}^{N}\sum_{m=1}^{M} I_{il}^{(1)} J_{jm}^{(1)} v_{lmk} - \frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} v_{imk}$$

$$+\frac{1}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M} I_{il}^{(1)} J_{jm}^{(1)} u_{lmk} \sum_{m=1}^{M} J_{jm}^{(1)} u_{imk} + \frac{1}{r_i}\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk} \sum_{m=1}^{M} J_{jm}^{(2)} u_{imk}\Bigg\}$$

$$+\frac{E}{2(1+\upsilon)}\Bigg\{\sum_{n=1}^{P} K_{kn}^{(2)} u_{ijn} + \sum_{l=1}^{N}\sum_{n=1}^{P} I_{il}^{(1)} K_{kn}^{(1)} w_{ljn} + \sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn} \sum_{l=1}^{N}\sum_{n=1}^{P} I_{il}^{(1)} K_{kn}^{(1)} u_{ljn}$$

$$+\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk} \sum_{n=1}^{P} K_{kn}^{(2)} u_{ijn}\Bigg\} = 0, \tag{A5}$$

$$\frac{E}{(1+\upsilon)(1-2\upsilon)r_i}\Bigg\{(1-\upsilon)\Bigg\{\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk} + \frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(2)} v_{imk} + \frac{1}{r_i^2}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk} \sum_{m=1}^{M} J_{jm}^{(2)} u_{imk}\Bigg\}$$

$$+\left\{\upsilon \sum_{l=1}^{N}\sum_{m=1}^{M} I_{il}^{(1)} J_{jm}^{(1)} u_{lmk} + \sum_{l=1}^{N} I_{il}^{(1)} u_{ljk} \sum_{l=1}^{N}\sum_{m=1}^{M} I_{il}^{(1)} J_{jm}^{(1)} u_{lmk}\right\}$$

$$+\upsilon\left\{\sum_{m=1}^{M}\sum_{n=1}^{P} J_{jm}^{(1)} K_{kn}^{(1)} w_{imn} + \sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn} \sum_{m=1}^{M}\sum_{n=1}^{P} J_{jm}^{(1)} K_{kn}^{(1)} u_{imn}\right\}\Bigg\}$$

$$+\frac{E}{2(1+\upsilon)}\Bigg\{\sum_{n=1}^{P} K_{kn}^{(2)} v_{ijn} + \frac{1}{r_i}\sum_{m=1}^{M}\sum_{n=1}^{P} J_{jm}^{(1)} K_{kn}^{(1)} w_{imn} + \frac{1}{r_i}\sum_{m=1}^{M}\sum_{n=1}^{P} J_{jm}^{(1)} K_{kn}^{(1)} u_{imn} \sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}$$

$$+\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk} \sum_{n=1}^{P} K_{kn}^{(2)} u_{ijn}\Bigg\} + \frac{E}{2(1+\upsilon)}\Bigg\{-\frac{1}{r_i^2}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk} + \frac{1}{r_i}\sum_{l=1}^{N}\sum_{m=1}^{M} I_{il}^{(1)} J_{jm}^{(1)} u_{lmk}$$

$$+\sum_{l=1}^{N} I_{il}^{(2)} v_{ljk} - \frac{1}{r_i}\sum_{l=1}^{N} I_{il}^{(1)} v_{ljk} + \frac{1}{r_i^2} v_{ijk} - \frac{1}{r_i^2}\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk} \sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}$$

$$+\frac{1}{r_i}\sum_{l=1}^{N} I_{il}^{(2)} u_{ljk}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}+\frac{1}{r_i}\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{l=1}^{N}\sum_{m=1}^{M} I_{il}^{(1)} J_{jm}^{(1)} u_{lmk}\Bigg\}$$
$$+\frac{E_i\frac{m_1}{a^{m_1}} r_i^{m_1-1}}{2(1+\upsilon)}\left\{\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}+\sum_{l=1}^{N} I_{il}^{(1)} v_{ljk}-\frac{1}{r_i} v_{ijk}+\frac{1}{r_i}\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\right\}$$
$$+\frac{E}{(1+\upsilon) r_i}\left\{\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}+\sum_{l=1}^{N} I_{il}^{(1)} v_{ljk}-\frac{1}{r_i} v_{ijk}+\frac{1}{r_i}\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\right\}=0, \tag{A6}$$

$$\frac{E}{(1+\upsilon)(1-2\upsilon)}\Bigg\{(1-\upsilon)\left\{\sum_{n=1}^{P} K_{kn}^{(2)} w_{ijn}+\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}\sum_{n=1}^{P} K_{kn}^{(2)} u_{ijn}\right\}$$
$$+\upsilon\left\{\sum_{l=1}^{N}\sum_{n=1}^{P} I_{il}^{(1)} K_{kn}^{(1)} u_{ljn}+\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{l=1}^{N}\sum_{n=1}^{P} I_{il}^{(1)} K_{kn}^{(1)} u_{ljn}\right\}+\upsilon\Bigg\{\frac{1}{r_i}\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}$$
$$+\frac{1}{r_i}\sum_{l=1}^{M}\sum_{n=1}^{P} J_{jm}^{(1)} K_{kn}^{(1)} v_{imn}+\frac{1}{r_i^2}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\sum_{m=1}^{M}\sum_{n=1}^{P} J_{jm}^{(1)} K_{kn}^{(1)} u_{imn}\Bigg\}\Bigg\}$$
$$-\frac{E\alpha}{(1-2\upsilon)}\sum_{n=1}^{P} K_{kn}^{(1)} T_{ijn}+\frac{E}{2(1+\upsilon)}\Bigg\{\sum_{l=1}^{N}\sum_{n=1}^{P} I_{il}^{(1)} K_{kn}^{(1)} u_{ljn}+\sum_{l=1}^{N} I_{il}^{(2)} w_{ljk}$$
$$+\sum_{l=1}^{N} I_{il}^{(2)} u_{ljk}\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}+\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{l=1}^{N}\sum_{n=1}^{P} I_{il}^{(1)} K_{kn}^{(1)} u_{ljn}\Bigg\}$$
$$+\frac{E_i\frac{m_1}{a^{m_1}} r_i^{m_1-1}}{2(1+\upsilon)}\left\{\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}+\sum_{l=1}^{N} I_{il}^{(1)} w_{ljk}+\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}\right\}$$
$$+\frac{E}{2(1+\upsilon) r_i}\Bigg\{\sum_{m=1}^{M}\sum_{n=1}^{P} J_{jm}^{(1)} K_{kn}^{(1)} v_{imn}+\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(2)} w_{imk}+\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(2)} u_{imk}\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}$$
$$+\frac{1}{r_i}\sum_{m=1}^{M} J_{jm}^{(1)} u_{imk}\sum_{m=1}^{M}\sum_{n=1}^{P} J_{jm}^{(1)} K_{kn}^{(1)} u_{imn}\Bigg\}$$
$$+\frac{E}{2(1+\upsilon) r_i}\left\{\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}+\sum_{l=1}^{N} I_{il}^{(1)} w_{ljk}+\sum_{l=1}^{N} I_{il}^{(1)} u_{ljk}\sum_{n=1}^{P} K_{kn}^{(1)} u_{ijn}\right\}=0. \tag{A7}$$

Governing equations of the steady-state temperature field for host and piezoelectric shells are defined as:

$$\frac{K_{\mathrm{FGM}}}{r_i}\sum_{l=1}^{N} I_{il}^{(1)} T_{ljk}+K_{\mathrm{FGM}}\sum_{l=1}^{N} I_{il}^{(2)} T_{ljk}+\left\{K_i\frac{m_3}{a^{m_3}} r_i^{m_3-1}\right\}\sum_{l=1}^{N} I_{il}^{(1)} T_{ljk}+K_{\mathrm{FGM}}\sum_{n=1}^{P} K_{kn}^{(2)} T_{ijn}=0, \tag{A8}$$
$$\frac{K_{r\mathrm{Piezo}}}{r_i}\sum_{l=1}^{N} A_{il}^{(1)} T_{ljk}+K_{r\mathrm{Piezo}}\sum_{l=1}^{N} A_{il}^{(2)} T_{ljk}+K_{z\mathrm{Piezo}}\sum_{n=1}^{P} C_{kn}^{(2)} T_{ijn}=0. \tag{A9}$$

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